

Liquidity and Information Aggregation in Large Markets^{*}

Paymon Khorrami[†]

João Thereze[‡]

July 20, 2026

Abstract

We study a financial market with information production and free entry of strategic traders. In contrast to Grossman and Stiglitz (1980), equilibrium prices may successfully aggregate information, and we provide necessary and sufficient conditions for full revelation. More generally, equilibrium information aggregation is determined by a sufficient statistic that depends on the information technology only through the marginal cost of information at the prior. Despite the market being large and competitive, microstructure frictions shape equilibrium outcomes: market liquidity encourages information acquisition. Unlike conventional competitive frameworks, which abstract from this mechanism, our model features an increasing relationship between noise and price informativeness, with implications for market design.

^{*}We thank Ehsan Azarmsa, Briana Chang, Christian Opp, Cecilia Parlatore, Adriano Rampini, Shri Santosh, Savi Sundaresan, Vish Viswanathan, and Chaojun Wang for helpful discussions.

[†]The Fuqua School of Business, Duke University — paymon.khorrami@duke.edu.

[‡]The Fuqua School of Business, Duke University — joao.thereze@duke.edu.

1 Introduction

How well do competitive financial markets aggregate information? The answer matters for central questions in economics, including the ability of decentralized markets to allocate resources efficiently. Unfortunately, according to the prevalent view, markets cannot efficiently aggregate information: if prices reveal information, investors may prefer to learn from prices rather than acquire information themselves (Grossman and Stiglitz, 1980). At the same time, it remains unclear how severely information aggregation fails, or which primitives determine the extent of the failure. We revisit the question of information aggregation from a different perspective and obtain two novel findings. Our analysis delivers both a *possibility* result—the wisdom of the crowd can hold—as well as a *precise characterization* from information technology to information aggregation.

Our starting point is a strategic trading model à la Kyle (1985) and Lambert et al. (2018). To these frameworks, we add two key ingredients: free entry, which is a hallmark of competition, and smooth information costs. Free entry implies that the number of traders grows large in equilibrium. Smooth information costs allow traders to be “informationally small.” Traders endogenously acquire information, but less so as entry makes markets increasingly competitive. Despite traders being informationally small, entry means that both the aggregate amount of information and price informativeness can grow arbitrarily large. We obtain necessary and sufficient conditions for information efficiency in the free-entry equilibrium. In cases where information aggregation is imperfect, we provide a precise characterization of price informativeness, and other market outcomes, entirely as a function of the marginal cost of the first piece of information.

Model and results. Our baseline model features a single risky asset traded by n ex-ante identical and risk-neutral strategic traders, as well as noise traders and an uninformed, competitive, market-maker. Strategic traders have access to an information technology allowing them to learn about the asset value at a cost. After acquiring information, strategic traders submit market orders, anticipating their price impact, and competitive market-makers set a break-even price upon observing the order flow. We are interested in the symmetric equilibrium with free entry, that is, when the number of traders is determined by a zero ex-ante profits condition. In our setting, the free-entry equilibrium corresponds to the large-market limit ($n \rightarrow \infty$) of this model. Although our baseline model studies risk-neutral traders who submit market orders (Kyle, 1985), our conclusions continue to hold when traders are risk-averse and submit limit orders (Kyle, 1989). These extensions preserve both the possibility of information aggregation and a precise

characterization of the degree of information efficiency.

In contrast with most of the strategic trading literature, we allow the information technology to be smooth, so that traders may decide to enter and acquire a small amount of information. Concretely, the information technology is represented by the cost of obtaining a signal with precision z above the prior, given by $c(z)$. In small markets, the details of the information technology matter to determine equilibrium. However, when the number of traders grows, the single critical object that characterizes price informativeness is $\chi := c'(0)$ —the marginal cost of precision at the prior.

Our first main result is that prices are fully informative if and only if $\chi = 0$. In a large market, each trader ends up acquiring a small amount of information z . Intuitively, this is a consequence of the strategic substitutability of information: when there are more traders who learn, prices are more revealing, reducing individual incentives for acquisition. When prices are arbitrarily revealing, information acquisition incentives are virtually absent. Therefore, it is immediate that a *necessary* condition for efficiency is that a small piece of information can be obtained at a vanishingly small marginal cost. The key contribution is to show this is also *sufficient*: as long as $\chi = 0$, then regardless of the other details of the cost function, prices become arbitrarily informative as the market grows. Although each individual trader acquires virtually no information, the aggregate level of information is high. The wisdom of the crowd obtains.

Our second main result precisely characterizes the large-market equilibrium for any $\chi > 0$. In general, we show that price informativeness is an increasing function of the following sufficient statistic:

$$\Gamma := \frac{1}{2} \left(\frac{\text{fundamental volatility} \times \text{noise trader volatility}}{\chi} \right)^2.$$

As Γ falls, information aggregation declines monotonically and ranges the full spectrum of price informativeness—from fully revealing to completely uninformative. The fact that the level of information is increasing in both fundamental volatility and noise is intuitive: both uncertainties allow strategic traders to “conceal” their information when trading, leading to greater information rents, thereby augmenting incentives for information acquisition. We then use our characterization of informativeness to obtain closed-form solutions for liquidity, price-volatility and trading volume.

As this discussion makes clear, our results depend critically on free entry and the smoothness of information costs. The extant literature obtained different results because it considered at most one of these assumptions at a time. For instance, Grossman and Stiglitz (1980) and Kyle (1989) study free-entry equilibria with fixed information costs,

while Verrecchia (1982) adopts smooth information costs but prevents free entry. (Surprisingly, agents make profits in the latter model, despite taking prices as given!)

The role of liquidity. In our model, strategic traders acknowledge market illiquidity. This is not the source of our results. On the contrary, traders acting strategically works against the possibility of information efficiency: price impact curbs aggressive trades, which mitigates the flow of information into the market. If traders (schizophrenically) were to disregard their price impact, they would impound more of their information into prices, making information efficiency more likely. Why, then, are informationally-efficient markets possible in our model? It turns out that markets with perfect information aggregation *are also* markets with perfect liquidity.

As a market becomes sufficiently informative, traders' incentives to deploy private information start decaying, such that the market's liquidity also begins improving. The improvement in liquidity continues to encourage information-acquisition, partially offsetting the dampening effect of better price informativeness. These effects appear transparently in a trader's marginal value of the first unit of precision, $V'(0)$, which is

$$V'(0) \propto (1 - \text{Informativeness})^2 \times \text{Liquidity}.$$

Thus, in an almost perfectly-informative market, it is the explosion in liquidity that sustains information acquisition.

We further highlight the role of liquidity by considering what happens if the entry of strategic traders is accompanied by noise. In particular, suppose that as n grows, noise trader volatility also grows without bound (this common assumption is used, e.g., in Hellwig, 1980; Verrecchia, 1982; Kyle, 1989; and Kovalenkov and Vives, 2014). The free-entry limit in this case is *always* one with perfect information aggregation, *for any marginal information cost* $\chi < \infty$. The reason: growing noise implies ever-improving liquidity, because market-makers are less wary of adverse selection. The improvement in liquidity boosts information-acquisition incentives. Ultimately, our results point to a tight link between perfect liquidity and information efficiency.

Conversely, under partial information aggregation, illiquidity issues do not vanish. In such cases where illiquidity persists, all of our comparative statics align with small-market strategic-trading models, even though our market is large and profits small. (This is not to say that all microstructure frictions survive. For example, we prove that the trading protocol—limit versus market orders—is irrelevant in our free-entry limit.)

These results imply a striking discrepancy between our model and Noisy Rational

Expectations Equilibrium (NREE) models. As the dominant paradigm, NREE models say that competitive financial markets are informationally-inefficient and perfectly liquid (at least from the perspective of the model’s traders, who take prices as given). Our model provides an entirely different view. Competitive markets can never be both inefficient and liquid: they are always *either* informationally-efficient *or* illiquid.

		Information efficiency	
		Imperfect	Perfect
Liquidity	Perfect	NREE	Our model
	Imperfect	Our model	

Implications for market design. We highlight a novel relationship between liquidity and informativeness which has market design implications. Formally, an increase in noise unambiguously raises both liquidity and informativeness. This result is in stark contrast with NREE, where the same increase in noise leads to *lower* informativeness. Furthermore, the discrepancy can be traced precisely to the improvement in liquidity, which has consequences in our model but not in NREE models. In other words, if liquidity were held fixed, price informativeness would fall with noise; but the improvement in liquidity is always enough to reverse the effect and raise informativeness.

By interpreting noise as retail order flow volatility, the result above has implications for market design. For example, markets for retail order flows are highly segmented (Dyhrberg et al., 2025; Ernst et al., 2023). A large share of retail orders flow to intermediaries called wholesalers, who may execute those orders outside of large exchanges. The typical justification is that wholesalers are willing to provide discounts to retail orders, which are largely uninformed. Our model suggests caution: removing retail order flow from the exchange removes noise, which lowers incentives for information production and may thereby reduce price informativeness. A conventional NREE-style model would conclude the exact opposite, that segmentation improves price efficiency (an example of such a result, whereby the presence of “dark pools” reduces on-exchange noise and thereby improves price discovery, can be found in Zhu, 2014).¹

¹Some theoretical research argues that such order flow segmentation may harm liquidity (Parlour and Rajan, 2003; Parlour and Seppi, 2003), but adverse consequences for informativeness are less explored.

Related literature. Our paper follows a long tradition of modeling strategic financial markets (Kyle, 1985, 1989; Lambert et al., 2018), especially a subset with endogenous information acquisition (Admati and Pfleiderer, 1988a, 1988b; Kyle, 1989; Subrahmanyam, 1991; Froot et al., 1992; Foster and Viswanathan, 1993; Peress, 2010; Lee, 2013; Glebkin and Kuong, 2023; Banerjee and Breon-Drish, 2020, 2022; Kacperczyk et al., 2025). Unlike this literature, we study the competitive limit of such models. Like us, Lee and Kyle (2018) also study the large-market limit of a strategic-trading model, albeit with exogenous information. They are concerned with what makes markets perfectly competitive, whereas we ask what makes markets efficient.²

Different to our approach, NREE models currently serve as the dominant paradigm for studying price efficiency in competitive markets. By taking free entry seriously and adopting smooth information costs, we arrive at an equilibrium that never looks like an NREE: prices are either highly informationally-efficient, or liquidity issues persist.

Three broad approaches have proved successful in circumventing the Grossman-Stiglitz paradox: adding a source of private value for information (Vives, 2011, 2014); preventing traders from conditioning trade on prices (Milgrom, 1981; Hellwig, 1982; Dubey et al., 1987); or explicitly modeling the price-formation process (Jackson, 1991). Our paper falls closest to the third tradition, despite adopting very different assumptions, with a novel emphasis on free entry and information costs.

Finally, we complement the research on information aggregation in strategic environments with common values, such as auctions (Jackson, 2003; Pesendorfer and Swinkels, 2000; Atakan and Ekmekci, 2023) and voting (Martinelli, 2006). Like these papers, we obtain conditions for aggregate levels of acquired information to be high even when individual incentives for learning vanish, but in a canonical financial market.

2 Model: Strategic Trading with Information Acquisition

We study a static strategic trading model with multiple potentially informed traders and costly information acquisition. Information acquisition is decided ex-ante, subject to a cost for signal precision. The baseline model below features risk-neutral traders who submit market orders to a competitive market-maker, following Kyle (1985) and Lambert

²Lee and Kyle (2018) define perfect competition more stringently than us: they require investors to submit identical demand curves as they would in NREE. They show that strategic markets become perfectly competitive in the large-market limit if and only if speculation incentives vanish relative to risk-sharing motives. Similarly, Kyle (1989) shows that a strategic model can only become perfectly competitive if the amount of noise also explodes. Others also bridging the gap between strategic agents and competitive markets include Holden and Subrahmanyam (1992), Vives (2002, 2011), and Kovalenkov and Vives (2014).

et al. (2018). Neither risk-neutrality nor the specific trading protocol are fundamental. In Section 3, we discuss how our results remain the same if trading occurs through limit orders instead of market orders, and how they extend to the case of risk-aversion.

A single risky asset with random payoff v is traded in a market with n risk-neutral strategic traders. In this market, there is also a representative uninformed market-maker and unmodeled noise traders that submit demand u . We assume that noise trader demand and fundamentals are independently normally distributed with mean normalized to zero: $v \sim \text{Normal}(0, \tau_v^{-1})$, $u \sim \text{Normal}(0, \tau_u^{-1})$, and $u \perp v$.

Before trading, investors have an opportunity to collect information about v in the form of a signal about the security value. Formally, strategic trader i has access to a signal $s_i = v + \varepsilon_i$, where ε_i is the error in the signal, which is normally distributed with variance τ_i^{-1} . At a cost, the investor can choose the variance of the error ε_i . We assume that trader's errors are independent of each other and fundamentals: $\varepsilon_i \perp \varepsilon_j$ for any pair of traders (i, j) , and $\varepsilon_i \perp v$.

The trading protocol follows Kyle (1985). Each trader submits a market order d_i , and noise trader demand u realizes. The market-maker observes aggregate demand $Y = \sum_i d_i + u$ and sets a break-even price given the information in the order flow:

$$p = \mathbb{E}[v|Y]. \quad (1)$$

If trader i demands d_i units of the asset and market prices are p , her ex-post profit is:

$$d_i(v - p).$$

Prior to trading, information is gathered subject to a cost. For convenience, we will let traders choose a normalization of information precision, $z_i \in [0, 1]$, which is defined as follows, using the variance of trader i 's error, τ_i^{-1} :

$$z_i := \frac{\tau_i}{\tau_i + 2\tau_v}. \quad (2)$$

If $z_i = 0$, then $\tau_i = 0$, and trader i 's signal is entirely uninformative. If $z_i = 1$, then $\tau_i = \infty$, and the signal is perfectly informative. Indeed, the precision of an agent's signal is a monotonic function of z_i .

Given a collection of precisions, $\mathbf{z} = (z_i)_{i=1}^n$, trader i chooses d_i to maximize expected profits, conditional on her signal realization s_i . Because traders are strategic, they anticipate that their order will change prices, given by a function $\mathbf{p}(\cdot, \mathbf{z})$ of the aggregate demand. Moreover, traders take as given the demand functions of other traders, $\mathbf{d}_j(\cdot, \mathbf{z})$,

which maps trader j 's private information to a market order. We let $\tilde{V}_i$ represent i 's profits given a collection of precisions conditional on a signal realization. The trader's problem is then:

$$\tilde{V}_i(s_i|\mathbf{z}) := \max_{d_i} \mathbb{E} \left[d_i \left(v - \mathbf{p} \left(\sum_{j \neq i} \mathbf{d}_j(s_j, \mathbf{z}) + d_i + u, \mathbf{z} \right) \right) \middle| s_i \right]. \quad (3)$$

Before the trading stage, trader i chooses how much information to collect while taking all other traders' precisions as given. That choice trades-off the benefits of information in the subsequent trading equilibrium against the cost of information acquisition:

$$\max_{z \in [0,1]} \mathbb{E} [\tilde{V}_i(s_i|z, \mathbf{z}_{-i})] - c(z), \quad (4)$$

where $c : [0,1] \rightarrow \mathbb{R}_+$ is a continuously differentiable and strictly convex function satisfying $c(0) = 0$, $c'(0) \geq 0$, and $c'(1-) = \infty$. The latter condition is simply for ease of exposition. Moreover, $\mathbf{z}_{-i}$ represents the vector of precisions of all other traders. We are now ready to define equilibrium.

Definition 1. An equilibrium is a collection of precisions, $\mathbf{z}$, demand functions $(\mathbf{d}_i)_{i=1}^n$, with $\mathbf{d}_i : \mathbb{R} \times [0,1]^n \rightarrow \mathbb{R}$, and a price function $\mathbf{p} : \mathbb{R} \rightarrow \mathbb{R}$, such that:

1. For any collection of precisions $\tilde{\mathbf{z}}$, $\{(\mathbf{d}_i(\cdot, \tilde{\mathbf{z}}))_{i=1}^n, \mathbf{p}(\cdot, \tilde{\mathbf{z}})\}$ is a linear trading equilibrium.

That is:

- (a) Traders optimize: for each trader i , given prices, other traders' demand functions, and signal realization s_i , demand $\mathbf{d}_i(s_i, \tilde{\mathbf{z}})$ solves (3).
- (b) Market-makers break even: $\mathbf{p}(Y, \tilde{\mathbf{z}}) = \mathbb{E}[v|Y]$,
- (c) Demand and prices are linear: there exist constants $\lambda(\tilde{\mathbf{z}}), \beta_i(\tilde{\mathbf{z}})$, such that: $\mathbf{p}(Y, \tilde{\mathbf{z}}) = \lambda(\tilde{\mathbf{z}})Y$, and $\mathbf{d}_i(s_i, \tilde{\mathbf{z}}) = \beta_i(\tilde{\mathbf{z}})s_i$.

2. $\mathbf{z}$ is an information equilibrium. That is, z_i solves (4) given $\mathbf{z}_{-i}$.

Regarding the notation above, we emphasize that $\mathbf{d}_i$ and $\mathbf{p}$ are used to as demand and pricing *functions*, which are defined for an arbitrary vector of precisions $\mathbf{z}$, whereas d_i and p are the equilibrium demand and price (not functions but random variables). While this notation is rather cumbersome in principle, most of the analysis below does not need it. Indeed, we show below that our notation substantially simplifies in terms of a few sufficient summary statistics.

After describing some basic properties of equilibrium for any finite n , most of the paper studies the *free-entry* or *competitive* limit of this market. As we will see, this free-entry limit is equivalent to the limit as the number of strategic traders to grow large ($n \rightarrow \infty$).

2.1 Linear trading equilibrium

For any information choice decision, $\mathbf{z}$, we can compute the equilibrium of the associated trading game. Although the result is a special case of Lambert et al. (2018), it is useful to express the equilibrium using the economic quantities we explore later. To that end, we define two critical statistics. Define *relative price informativeness* as the fraction of the fundamental variance left after observing prices in an equilibrium:³

$$I(\mathbf{z}) := 1 - \frac{\text{Var}[v|p, \mathbf{z}]}{\text{Var}[v]}.$$

As $I(\mathbf{z}) \rightarrow 1$, prices become perfectly informative, whereas $I(\mathbf{z}) \rightarrow 0$ corresponds to zero information revealed by prices. Moreover, we measure *liquidity* as the inverse of the price impact of a marginal noise trade, which is the inverse of Kyle's lambda:

$$L(\mathbf{z}) := \frac{1}{\lambda(\mathbf{z})}.$$

Our first result characterizes the unique linear trading equilibrium through these two variables and trader i 's value function.

Lemma 1. *Fix any $\mathbf{z}$. There is a unique linear trading equilibrium. In that equilibrium, informativeness, liquidity, and trader i 's value are given by*

$$\begin{aligned} I(\mathbf{z}) &= \frac{\mathbf{z} \cdot \mathbf{1}}{1 + \mathbf{z} \cdot \mathbf{1}} \\ L(\mathbf{z}) &= \sqrt{\frac{2\tau_v}{\tau_u}} \frac{1 + \mathbf{z} \cdot \mathbf{1}}{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}} \\ V_i(\mathbf{z}) &:= \mathbb{E}[\tilde{V}_i(s_i, \mathbf{z})] = \frac{1}{2\tau_v} (1 - I(\mathbf{z}))^2 L(\mathbf{z}) (z_i + z_i^2) \end{aligned}$$

Define the *aggregate information* in the market as the sum of all normalized precisions, $\mathbf{z} \cdot \mathbf{1}$. Lemma 1 shows that more information in the market mechanically increases price informativeness. Moreover, liquidity depends not only on the level, but also on the

³Many definitions of price-informativeness have been used in the literature. We follow the usage in Grossman and Stiglitz (1980).

dispersion of information: when information is more equally shared among traders, the market is more liquid. The relationship between liquidity and the level of aggregate information is more subtle and, in fact, non-monotonic. When there is no information in the market, liquidity is abundant as there is low adverse selection and the market-maker cannot infer anything from the order flow. The price is then fixed at the prior mean, 0, and insensitive to changes in the order flow. On the other extreme, when aggregate information is infinite, liquidity is again plentiful as the price is then pinned down to the fundamental and incentives to trade on information are virtually non-existent.

Trader i 's indirect utility function conveys three economic forces. First, the individual trader benefits from being better informed, as she can use that information to submit informed market orders, which are not perfectly detected by the market-maker due to noise trader demand. Therefore, there is a direct, positive effect of z_i on V_i . Second, a higher informativeness reduces the marginal payoff of information, because equilibrium prices are closer to the real valuation of the asset. Finally, and important for our paper, liquidity also affects the value of information: a deeper market allows traders to leverage their private knowledge more, as prices do not move as much against their own trades. That means that the value of information is larger in deeper markets.

2.2 Information equilibrium and free entry

Next, we describe the information equilibrium, where traders choose $z_i \in [0, 1]$ to solve (4). We focus on a symmetric equilibrium, in which $\mathbf{z} = z^n \mathbf{1}$ for some scalar z^n , which depends on the number of traders in the market, n . We will let p^n and Y^n denote the random variables that represent equilibrium prices and aggregate order flow, respectively, in this symmetric equilibrium with n traders. As it will be very important in the remainder of the analysis, let us define

$$\chi := c'(0),$$

the marginal cost of information at the prior—i.e. the cost of the first unit of precision.

At a symmetric information equilibrium, we must have

$$\left. \frac{dV_i(\mathbf{z})}{dz_i} \right|_{\mathbf{z}=z^n \mathbf{1}} \leq c'(z^n), \quad \text{with equality if } z^n > 0. \quad (5)$$

Therefore, the fundamental quantity determining equilibrium is the marginal value of

information. From [Lemma 1](#), one can show:

$$\frac{dV_i(z_i, \mathbf{z}_{-i})}{dz_i} = \frac{1}{2\tau_v} (1 - I(\mathbf{z}))^2 L(\mathbf{z}) (1 + 2z_i) + o(\mathbf{z}) \leq \frac{1}{2\tau_v} (1 - I(\mathbf{z}))^2 L(\mathbf{z}) (1 + 2z_i), \quad (6)$$

where $o(\cdot)$ is a negative function such that $o(\mathbf{z}) \xrightarrow{z_i \rightarrow 0} 0$ when $\mathbf{z}_{-i} > 0$, and $o(\mathbf{z}\mathbf{1}) \xrightarrow{z \rightarrow 0} 0$. In other words, as a single trader becomes informationally small, their marginal value of information approximately coincides with the marginal value that ignores their equilibrium impacts on aggregate informativeness and liquidity.

For any fixed size- n economy, there is an information equilibrium with positive information acquisition, as long as $\chi := c'(0) < \infty$. Indeed, notice that liquidity $L(\mathbf{z})$ grows without bound as all agents become uninformed. In that case, (6) shows that $\lim_{z \rightarrow 0} \frac{dV_i(\mathbf{z})}{dz_i} \Big|_{\mathbf{z}=\mathbf{z}\mathbf{1}} = \infty$, so traders are always willing to pay some cost to acquire a little bit of information. Hence, (6) holds with equality.

The remainder of the paper studies the *free-entry* or *competitive* limit of this market. It turns out that this is equivalent to the limit as the number of strategic traders to grow large ($n \rightarrow \infty$). Indeed, traders always make profits net of information costs for every finite n . To see this, notice from [Lemma 1](#) that $V_i(\mathbf{0}) = 0$ when n is finite. Additionally, recall the properties of information costs—that $c(\cdot)$ is a convex function satisfying $c(0) = 0$. Since $z^n > 0$ in any symmetric equilibrium, as shown above, we thus have that $V_i(z^n \mathbf{1}) - c(z^n) > 0$. This shows that the free-entry limit requires $n \rightarrow \infty$. Conversely, we will argue after the next result that trader profits vanish as $n \rightarrow \infty$, so that free entry is exhausted in this limit.

We start by establishing a preliminary but fundamental result that each trader must become “informationally small” as the market grows.

Lemma 2. *Individual information-gathering vanishes asymptotically in a symmetric equilibrium: $z^n \xrightarrow{n \rightarrow \infty} 0$.*

[Lemma 2](#) results from the strategic substitutability in information-gathering. If z^n does not converge to zero, then the aggregate amount of information in the market must be growing without bound. But if that is the case, it becomes impossible for individual traders to disguise their demand among noise, which drives the benefit of information acquisition to zero, contradicting z^n bounded away from zero.

An implication of informationally-small traders is that profits necessarily vanish as $n \rightarrow \infty$. Indeed, using the expressions in [Lemma 1](#) and the limit result of [Lemma 2](#), we

have that

$$V_i(z^n \mathbf{1}) - c(z^n) = \sqrt{\frac{1}{2\tau_v\tau_u} \frac{\sqrt{1+z^n}}{1+nz^n} \frac{\sqrt{z^n}}{\sqrt{n}}} - c(z^n) \xrightarrow{n \rightarrow \infty} 0 \quad (7)$$

Consequently, the free-entry limit is, in fact, equivalent to the requirement that $n \rightarrow \infty$.

Thus, we have a first insight behind our main results and how they depend on our two novelties: free entry and smooth information costs. Free entry, a hallmark of competition, is equivalent to a large number of traders. With smooth information costs, these large number of traders will necessarily be informationally-small, according to [Lemma 2](#). Ultimately, this suggests that the information cost only matters in equilibrium via its local derivative at the prior, i.e., via $\chi = c'(0)$.

2.3 Information aggregation

We are interested in price informativeness in the competitive limit, namely as the number of traders, n , grows to infinity.

Definition 2. *Competitive price informativeness is⁴*

$$\mathcal{I} := \lim_{n \rightarrow \infty} I(z^n \mathbf{1}).$$

We now present a principal result of the paper.

Theorem 1. *Competitive informativeness $\mathcal{I} \in [0,1]$ is the unique solution to:*

$$\frac{\mathcal{I}}{(1 - \mathcal{I})^3} = \Gamma, \quad (8)$$

$$\text{where } \Gamma := \frac{1}{2} \frac{1}{\chi^2 \tau_u \tau_v}. \quad (9)$$

[Theorem 1](#) characterizes the asymptotic level of informativeness in the market as a function of three primitives: noise trader uncertainty, τ_u^{-1} , fundamental uncertainty, τ_v^{-1} , and the marginal cost of the first piece of information, χ . Prices naturally are more informative when the marginal cost of information at the prior, χ , is lower. Moreover, prices are more informative either when fundamental uncertainty or noise uncertainty are higher. This characterization is simple, clean, and relatively detail-free with respect to the information cost function, depending only on its derivative at the prior mean, χ . As information costs are difficult to observe, this detail-free characterization of the level

⁴In the proof of [Theorem 1](#), this limit is shown to exist.

of information is advantageous, compared to expressions that rely heavily on the shape of $c(\cdot)$ —which arise in the case of NREE models.

The first major implication of [Theorem 1](#) is that price informativeness can take any value, including arbitrarily high values, in stark contrast with NREE models (e.g., Grossman and Stiglitz, 1980). Indeed, as χ varies from 0 to ∞ , the sufficient statistic Γ spans the whole real line. Furthermore, arbitrarily high price informativeness can obtain even when it is very expensive to produce a precise signal, so long as the first piece of information is relatively cheap—that is, $\chi = c'(0)$ is all that matters.⁵

Corollary 1. *Prices are asymptotically fully-revealing—i.e. $\mathcal{I} = 1$ —if and only if $\chi = 0$.*

Why is $\chi = 0$ necessary and sufficient for price efficiency? First, the necessity of $\chi = 0$ is straightforward: as the level of aggregate information goes to infinity, information swamps the effect of noise and traders cannot hide their informed trades anymore, leading the value of information to decline to zero. If marginal costs of information at the prior were positive, this would eventually lead any particular individual to acquire exactly zero information, which could not lead to fully-revealing prices. On the other hand, sufficiency of $\chi = 0$ is due intuitively to the following: if price informativeness is imperfect, then trading on private information is profitable. With positive profits and zero cost of the first piece of information, traders would always want to learn a strictly positive amount. With the number of traders growing to infinity, due to free entry, aggregate information is thus unbounded.

Thus, when the marginal cost of the first piece of information is zero, markets perfectly aggregate dispersed information, despite information being endogenous. This is in line with recent research on information aggregation in common-value auctions, which also finds that dispersed information can be well-reflected in prices if and only if the marginal cost of information is zero at the prior (Atakan and Ekmekci, 2023).

2.4 The role of liquidity

Our starting point is a strategic model, in which traders internalize their price impact. But we also study the large-market limit of such a model. Do liquidity frictions automatically vanish in this limit? If not, under what conditions does illiquidity persist?

⁵[Appendix D](#) studies a version of such an NREE with endogenous information and smooth costs, as in Verrecchia (1982). There, we show that $\mathcal{I} = 1$ generically cannot arise. Instead, informativeness is bounded away from one even if the information cost function satisfies $c'(0) = 0$. A key reason is that traders collect positive amounts of information, $z^* > 0$, which means that the entire cost function $c(z)$ matters—not just the local derivative $c'(0)$. Unless this entire cost function vanishes, informativeness stays bounded.

Proposition 1. *Competitive liquidity $\mathcal{L} := \lim_{n \rightarrow \infty} L(z^n \mathbf{1})$ satisfies*

$$\mathcal{L} = \sqrt{\frac{2\tau_v}{\tau_u}} \frac{1}{\sqrt{\mathcal{I}(1 - \mathcal{I})}} = \frac{2\tau_v\chi}{(1 - \mathcal{I})^2}$$

An immediate implication of [Proposition 1](#) is that $\mathcal{L} = \infty$ when $\mathcal{I} = 1$. A market with fully-revealing prices is also a market with perfect liquidity. Why? As strategic traders collect more information in the aggregate, they simultaneously become informationally-small at the individual level and therefore submit smaller orders individually. This implies two things. First, the large amount of aggregate information means that overall order flow becomes a near perfect signal for the fundamental value. Second, small strategic-trader orders means that market-makers can be confident that they will not be adversely selected. Because of these two forces, market-makers are then able to set prices approximately equal to the fundamental value and absorb any hypothetical order flow—i.e., markets are both highly informative and perfectly liquid.

The explosion in liquidity in fact sustains information-acquisition. A simple way to see this is to revisit the marginal value of information in equation (6) but in the large-market limit:

$$\lim_{n \rightarrow \infty} \left. \frac{dV_i(z_i, \mathbf{z}_{-i})}{dz_i} \right|_{\mathbf{z} = z^n \mathbf{1}} = \frac{1}{2\tau_v} (1 - \mathcal{I})^2 \mathcal{L}. \quad (10)$$

If informativeness becomes large in a market with bounded liquidity ($\mathcal{I}$ large and $\mathcal{L}$ bounded), incentives to acquire information vanish quickly. If so, informativeness could never become perfect. Roughly speaking, the fact that liquidity grows without bound helps explain why information can be fully aggregated.

On the other hand, when prices are only partially-revealing, liquidity is necessarily imperfect. That is, $\mathcal{I} < 1$ corresponds to $\mathcal{L} < \infty$.⁶ In other words, liquidity frictions can persist even in the large-market limit. Combining these results, our competitive equilibrium has *either* (i) a perfectly informative price *or* (ii) an imperfectly liquid market. Accordingly, our equilibrium never coincides with conventional frameworks like NREE which can never have (i) or (ii): an NREE essentially assumes perfect liquidity (no trader ever has price impact) and can never feature fully-informative prices.

When liquidity is imperfect, our large, competitive market looks instead like a smaller, thinner market. A clear manifestation of this phenomenon is the reaction of markets to noise. A direct corollary of [Theorem 1](#) and [Proposition 1](#) is that price informativeness

⁶This is except for the knife-edge scenario where prices are fully uninformative— $\mathcal{I} = 0$ —which occurs when traders acquire no information (e.g., if $\chi = \infty$).

and liquidity always grow with noise.

Corollary 2. *Informativeness and liquidity grow with noise, i.e., $\mathcal{I}$ and $\mathcal{L}$ are increasing in τ_u^{-1} .*

Remarkably, [Corollary 2](#) conflicts with what is obtained in NREE models. Indeed, while liquidity increases in noise in both models, informativeness is always decreasing in noise in NREE models.⁷ [Figure 1](#) displays an example. Because of this difference, examining how markets respond to variation in noise can potentially distinguish the frameworks empirically.

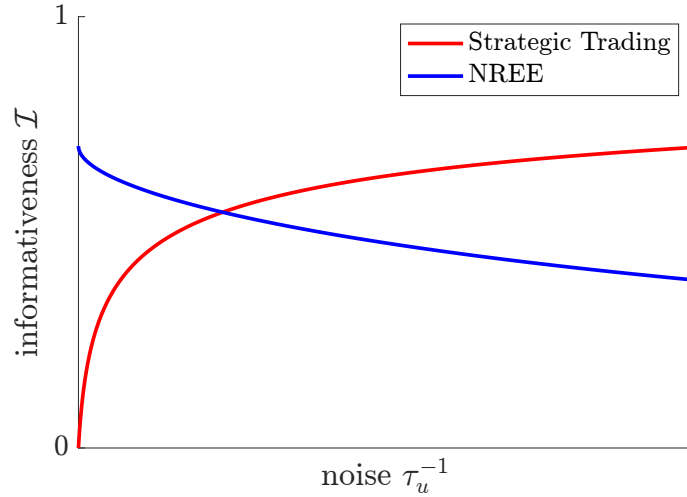


Figure 1: Informativeness against noise, in our strategic-trading model and the NREE. Baseline parameters: $\tau_u^{-1} = 0.5$ and $\tau_v^{-1} = 0.5$. For the information cost functions, $\chi = 0.1$ in both models. In the NREE, we use the cost function $c(z) = \chi(\frac{z}{1-z})$ which satisfies $c'(0) = \chi$. The NREE also has a risk aversion parameter, which we set to $\gamma = 3$. See [Appendix D](#) for more details on the NREE solution.

The overarching reasoning behind [Corollary 2](#) is that liquidity plays a role in trader's behavior. To be concrete, an increase in noise-demand uncertainty has three effects on equilibrium informativeness. The first effect, present in both NREE models and in strategic trading models, is direct: an increase in noise traders makes prices mechanically less informative. The second effect is that this reduction in informativeness increases the marginal value of information by creating profitable opportunities—deviations between the price and the fundamental value—which lead to an increase in information acquisition. This force, also present in NREE, is never enough to compensate for the mechanical deterioration of informativeness.

The third force—a liquidity effect—reverses that deterioration, leading to higher informativeness. An increase in noise deepens the market, making the trader's *marginal* trade more profitable, as prices move less against her. Because she is able to trade

⁷Our [Appendix D](#) considers an NREE with endogenous information, à la Verrecchia (1982). We show that, for any valid information cost function, $\mathcal{I}$ is *always decreasing* in τ_u^{-1} . See [Proposition D.1](#), part (i).

more aggressively and make more profits, the trader is willing to acquire even more information. The positive impact of liquidity on information-acquisition incentives is clearly visible in equation (10). On aggregate, prices become more informative precisely because the market becomes more liquid. This channel was noted by Admati and Pfleiderer (1988b) and is consistent with empirical evidence (Collin-Dufresne and Fos, 2015). The novelty in our paper is that the positive co-movement between liquidity and informativeness continues to survive even in large, competitive markets.⁸

This distinction between our model and NREE is relevant for several applications. One example is the increasing routing of retail order-flow to off-exchange venues. To the extent that a large part of retail volume is executed off-exchange, the uncertainty of on-exchange noise trading is reduced. NREE suggests that this segmentation brings no trade-offs, because exchange prices should become more informative, and retail traders benefit from better deals with wholesalers. Our model suggests caution: the decrease in noise should deteriorate the information quality on the exchange. This deterioration is difficult to detect simply by comparing on-exchange and off-exchange pricing, not only because off-exchange transactions can be required to occur at “better” prices than on-exchange transactions (e.g., Regulation NMS), but more importantly because the counterfactual without segmentation is unobserved. Our model suggests that an integrated market would have more informative prices both on- and off-exchange.

Information aggregation with growing noise. The assumption that noise variance, τ_u^{-1} , is constant raises a natural concern. Perhaps perfect information aggregation is driven by scale alone: as the market grows, many mildly informed traders simply overwhelm a fixed amount of noise. To address this concern, we let the noise variance in a market with n traders be $\tau_{u,n}^{-1}$ and increasing in n . If this sequence of variances converges to a finite limit as the number of traders grows, our previous results apply. We therefore focus on the case in which noise variance diverges and characterize equilibrium for any rate of growth.⁹

Although natural, this concern is misguided. It is true that more noise makes prices less informative, holding information fixed. Yet it also deepens liquidity, as discussed above, which raises the return to becoming informed. With free entry, this liquidity

⁸Another explanation for the positive correlation between adverse selection and liquidity is that informed investors time the market to trade when liquidity is higher (Collin-Dufresne and Fos, 2016).

⁹In fact, the assumption that noise grows with the number of traders often imposed and is often an important assumption on the way to obtaining a competitive limit (see Hellwig, 1980; Verrecchia, 1982; Kyle, 1989; and Kovalenkov and Vives, 2014). For instance, in Hellwig (1980) and Verrecchia (1982), the amount of noise variance grows at the rate of n^2 , because each trader is assumed to enter with a random supply of shares whose variance rises at a rate of n .

effect is powerful. Indeed, when noise volatility grows without bound, regardless of its growth rate, information efficiency is not only possible; it is required.

Proposition 2. Suppose $\chi < \infty$. If $\tau_{u,n}^{-1} \rightarrow \infty$, then $\mathcal{I} = 1$ and $\mathcal{L} = \infty$. Moreover, a free-entry limit exists if $\limsup_{n \rightarrow \infty} \frac{\tau_{u,n}^{-1}}{n^3} = 0$.

2.5 Volatility and Volume

In this section, we explore additional market observables—volatility and volume—and compare their behavior to their NREE counterparts. These observables may be of interest because they are easy to identify from market data, unlike liquidity and especially informativeness. As a result, the end of this section discusses how volatility and volume may actually be used, joint with the model's equations, to identify informativeness.

Definition 3. Price volatility $\mathcal{V}$ is the standard deviation of the asymptotic price, i.e.,

$$\mathcal{V} := \text{Std} \left[\lim_{n \rightarrow \infty} p^n \right]$$

Definition 4. Trading volume $\mathcal{Y}$ is the expected asymptotic absolute dollar demand:

$$\mathcal{Y} := \mathbb{E} \left[\left| \lim_{n \rightarrow \infty} p^n Y^n \right| \right]$$

Proposition 3. In the competitive limit, prices converge in distribution to:

$$\lim_{n \rightarrow \infty} p^n = \mathcal{I} v + \sqrt{\tau_v^{-1} \mathcal{I} (1 - \mathcal{I})} \eta,$$

for some $\eta \sim \mathcal{N}(0,1)$, such that $\eta \perp v$. In addition

$$\begin{aligned} (\text{volatility}) \quad \mathcal{V} &= \sqrt{\frac{\mathcal{I}}{\tau_v}} \\ (\text{volume}) \quad \mathcal{Y} &= \sqrt{\frac{2}{\tau_u \tau_v} \frac{\mathcal{I}}{1 - \mathcal{I}}} \end{aligned}$$

Under $\chi = 0$, we already found that $\mathcal{I} = 1$ (fully informative prices) and $\mathcal{L} = \infty$ (perfect liquidity). In addition, the results of [Proposition 3](#) say that $\mathcal{V} = \tau_v^{-1/2}$ (maximal volatility) and $\mathcal{Y} = \infty$ (infinite volume). These results hold regardless of the amount of fundamental uncertainty τ_v^{-1} and the amount of noise τ_u^{-1} .

Next, consider the case of partial efficiency: $\chi > 0$. As already shown in [Corollary 2](#), informativeness is increasing in noise, which is opposite to the NREE comparative

static. For this reason, informativeness and the other market observables can co-move oppositely in our framework, relative to an NREE.

A particularly stark example is the volatility-informativeness relation. In our model, volatility is always increasing in informativeness, no matter which parameter is moving (τ_u , τ_v , or χ)—this is clear by examining the expression $\mathcal{V} = \sqrt{\tau_v^{-1} \mathcal{I}}$ and noting from [Theorem 1](#) that $\mathcal{I}$ is decreasing in τ_u , τ_v , and χ . By contrast, in an NREE model, volatility is almost-always decreasing in informativeness in a sense that we make precise in the appendix.¹⁰ [Figure 2](#) illustrates an example.

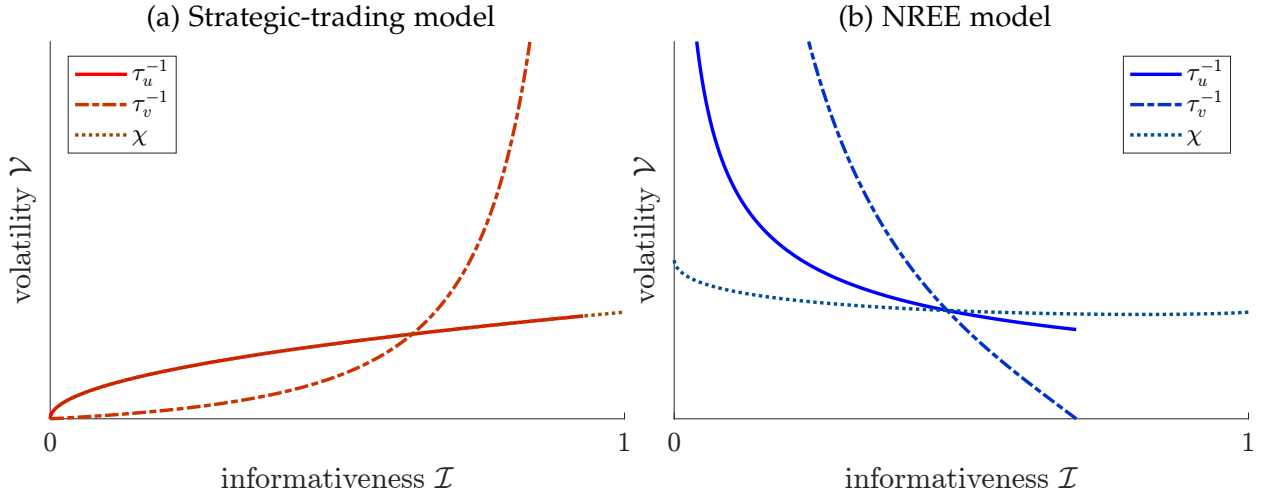


Figure 2: Informativeness against Volatility, in our strategic-trading model and the NREE. We vary the parameters τ_u^{-1} , τ_v^{-1} , and χ one at a time. The baseline parameters and NREE cost function $c(z) = \chi(\frac{z}{1-z})$ are the same as in [Figure 1](#).

Of interest is the nuanced impact of fundamental uncertainty τ_v^{-1} . As fundamental uncertainty rises, overall price volatility always rises in both models. At the same time, higher uncertainty induces two opposing forces pertinent to information-acquisition. First, with more risk in the economy, risk-averse traders become more hesitant to trade and collect information. Second, with more uncertainty, there is more potential information to be collected. In NREE models, the first effect always dominates the second, and informativeness is always falling in fundamental uncertainty (this is proved in [Proposition D.1](#)). In our baseline strategic-trading model, the risk effect is zero because traders are risk-neutral, so informativeness is always rising in fundamental uncertainty (this is a direct corollary of [Theorem 1](#)). Interestingly, our extended model with risk aversion, described in [Section 3](#), preserves the same pattern. The reason traces to a deep feature of our competitive equilibrium—that traders are informationally-small and therefore trade small amounts, implying that risk effects are always second-order.

¹⁰This is shown in [Appendix D, Proposition D.1](#), by comparing parts (i) and (iii).

Finally, given the observables in our model behave differently than in NREE models, it stands to reason that price informativeness should inherit novel dynamics. To open an avenue for research into this issue, [Appendix A.7](#) shows how to identify price informativeness within our model, solely using observations of volatility and volume. The identification can be done at high frequency and contrasts methodologically with accounting-based approaches in the literature (see Kothari and Sloan, [1992](#); Bai et al., [2016](#); Kacperczyk et al., [2021](#); Farboodi et al., [2022](#); Dávila and Parlato, [2023](#), [2025](#)).

3 Discussion of Assumptions

In this section we discuss and extend the key assumptions of our baseline model. First, we show that modeling the trading protocol as one in which traders submit limit orders (rather than market orders) is irrelevant for our results. In particular, all of our results remain unchanged when traders submit limit orders, as long as there is free entry of uninformed speculators (which serves as the counterpart to an uninformed market maker). Second, we consider risk-averse traders and show that the results still hold qualitatively: informativeness is determined by the same sufficient statistic, but now we need one more equation to determine the total amount of information in the market (which matters for the level of liquidity). We provide a characterization of this pair of equations, extending [Theorem 1](#), and obtain the relevant market measures. We show that our central comparative statics also hold in this setting. Finally, we discuss the role of smooth information costs versus fixed costs.

Limit Orders. Consider the same setup as in our baseline model, with two modifications. First, the trading protocol is as in Kyle ([1989](#)): traders submit limit orders, in the form of a demand schedule, and markets clear. Second, for any fixed level of informed traders, n , there is free entry of uninformed traders. The second assumption guarantees that there is sufficient uninformed demand to arbitrage away all the public information. In particular, if one maintains the same definition of Y as the aggregate order-flow from informed and noise traders, free entry of uninformed traders keeps prices unbiased: $p = \mathbb{E}[v|Y]$, thereby serving a role analogous to our baseline model’s market maker. In this setup, [Theorem 1](#), [Proposition 1](#), and [Proposition 3](#) hold exactly as stated. We do not formalize this claim—it is a direct consequence of [Theorem 2](#) below. A key implication of is that the trading protocol becomes irrelevant in competitive markets: although some microstructure phenomena persist in large markets, others do not.

Risk-Aversion. Next, maintaining the assumption of limit orders and free entry of uninformed investors, we also model traders as risk-averse. Traders are assumed to be CARA utility maximizers with risk-aversion γ . That is, the ex-post payoff for trader i from demanding d_i at price p is:

$$\frac{1}{\gamma} \left(1 - e^{-\gamma(v-p)d_i} \right).$$

We develop this model in [Appendix B](#). Compared to a model with risk-neutral traders, the key difference is that the amount of the acquired information that gets revealed by prices now varies with fundamentals. When traders are risk-neutral, as in Kyle (1985), exactly 50% of the private information held by traders is leaked through prices. Under risk-aversion, this fraction is $\varphi \leq \frac{1}{2}$, which plays a central role in the equilibrium characterization. Formally, define:

Definition 5. Price leakage φ is the fraction of all the information acquired which gets revealed by prices:¹¹

$$\varphi := \lim_{n \rightarrow \infty} \frac{\text{Var}[v|p, \{s_i\}_{i=1}^n]}{\text{Var}[v|p]}.$$

Theorem 2. Assume $\gamma\chi < 1$. In a model with risk-averse traders, limit orders and free entry of uninformed traders, competitive informativeness $\mathcal{I}$ is the unique solution to:

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3} \frac{1-\varphi}{2\varphi^2} = \Gamma,$$

where $\varphi = \frac{1}{2} - \frac{1}{2} \frac{\gamma\chi}{1-\mathcal{I}}$. Moreover, liquidity satisfies:

$$\mathcal{L} = \sqrt{\frac{\tau_v}{\tau_u(1-\varphi)}} \frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})}}.$$

Finally, $\mathcal{I}$ is increasing with τ_u^{-1} .

If $\gamma\chi \geq 1$, there is no information acquisition in equilibrium.

[Theorem 2](#) shows that our results obtained in the risk-neutral model are not knife-edge. First, when $\chi = 0$, prices become fully-revealing, and the equations above look exactly as they do in the risk-neutral model. Second, when $\chi > 0$, there is an interior level of price informativeness which, crucially, is still increasing in noise.

¹¹Lee and Kyle (2018) call this fraction *relative informational efficiency*.

These results are complicated, in principle, by the price leakage variable φ . When $\chi > 0$, price leakage φ is now less than one-half and thus less than the risk-neutral model. Furthermore, one can show that φ decreases with noise, which serves to reduce the amount of information revealed to the market. Nevertheless, informativeness $\mathcal{I}$ still always rises with noise, because the total amount of information goes up so sharply that it more than compensates the decline in φ .

Fixed information costs. For comparison, [Appendix C](#) studies a version of our baseline model with fixed information costs. In that case, it is well-understood that an equilibrium with a large quantity of aggregate information *can only emerge if the entire cost function vanishes* (Kyle, 1989; Vives, 2014). This stands in stark contrast to our result with smooth costs. The intuition for this result is that with fixed costs, the number of informed traders stabilizes as the market becomes large, and so the quantity of aggregate information stabilizes too. The remaining qualitative results hold: there is a quite similar characterization of price-informativeness and liquidity, and price-informativeness is again increasing in noise-trader volatility—i.e., decreasing in τ_u .

4 Conclusion

We examine information efficiency in the large-market limit of a strategic trading model. Our main contribution is an explicit characterization of the mapping between information technology and aggregate information. In particular, if the marginal cost of information is zero at the prior, then aggregate information is infinitely-large, prices are efficient, and markets are perfectly liquid. We think of this result as restoring the “wisdom of the crowd” in competitive markets, because it captures the well-understood potential for prices to aggregate dispersed information, but in a setting with endogenous information acquisition.

We go on to characterize additional differences between our framework and the literature on Noisy Rational Expectations Equilibria (NREE). The starkest result is that price informativeness in the two classes of models respond oppositely to noise. This discrepancy can be traced to a liquidity effect, whereby noise deepens markets and creates incentives to acquire information. As such, we highlight theoretically that conventional ways of assuming perfect competition are not innocuous. Practically, we highlight some implications for market design: if noise can be proxied by retail trade, then structures that segment or dampen retail trade can, perhaps surprisingly, be inefficient.

While our model abstracts from many important and realistic features, our aim is not

to explain a host of empirical patterns at this stage. Instead, we revisit the foundations. Our model should be viewed as a new competitive benchmark that is distinct—both positively and normatively—from the conventional NREE benchmark in which investors take prices as given. To address richer patterns in competitive markets, future research could extend our framework to include such features as sentiment, learning about noise, risk-sharing considerations, asymmetries between large and small traders, and the like. Importantly, our framework, being simple and tractable, can serve as a viable foundation atop which these additional features can be built.

References

- Admati, A. R., and Pfleiderer, P. (1988a). Selling and trading on information in financial markets. *The American Economic Review*, 78(2), 96–103.
- Admati, A. R., and Pfleiderer, P. (1988b). A theory of intraday patterns: Volume and price variability. *The Review of Financial Studies*, 1(1), 3–40.
- Atakan, A. E., and Ekmekci, M. (2023). *Information aggregation in auctions with costly information* (tech. rep.). Working paper.
- Bai, J., Philippon, T., and Savov, A. (2016). Have financial markets become more informative? *Journal of Financial Economics*, 122(3), 625–654.
- Banerjee, S., and Breon-Drish, B. (2020). Strategic trading and unobservable information acquisition. *Journal of Financial Economics*, 138(2), 458–482.
- Banerjee, S., and Breon-Drish, B. (2022). Dynamics of research and strategic trading. *The Review of Financial Studies*, 35(2), 908–961.
- Bernhardt, D., and Taub, B. (2006). Kyle v. kyle ('85 v.'89). *Annals of Finance*, 2(1), 23–38.
- Collin-Dufresne, P., and Fos, V. (2015). Do prices reveal the presence of informed trading? *The Journal of Finance*, 70(4), 1555–1582.
- Collin-Dufresne, P., and Fos, V. (2016). Insider trading, stochastic liquidity, and equilibrium prices. *Econometrica*, 84(4), 1441–1475.
- Dávila, E., and Parlatore, C. (2023). Volatility and informativeness. *Journal of Financial Economics*, 147(3), 550–572.
- Dávila, E., and Parlatore, C. (2025). Identifying price informativeness. *The Review of Financial Studies*, hhaf051.
- Dubey, P., Geanakoplos, J., and Shubik, M. (1987). The revelation of information in strategic market games: A critique of rational expectations equilibrium. *Journal of Mathematical Economics*, 16(2), 105–137.

- Dyhrberg, A. H., Shkilko, A., and Werner, I. M. (2025). The retail execution quality landscape. *Journal of Financial Economics*, 168, 104051.
- Ernst, T., Malenko, A., Spatt, C., and Sun, J. (2023). What does best execution look like?
- Farboodi, M., Matray, A., Veldkamp, L., and Venkateswaran, V. (2022). Where has all the data gone? *The Review of Financial Studies*, 35(7), 3101–3138.
- Foster, F. D., and Viswanathan, S. (1993). The effect of public information and competition on trading volume and price volatility. *The Review of Financial Studies*, 6(1), 23–56.
- Froot, K. A., Scharfstein, D. S., and Stein, J. C. (1992). Herd on the street: Informational inefficiencies in a market with short-term speculation. *The Journal of Finance*, 47(4), 1461–1484.
- Glebkin, S., and Kuong, J. C.-F. (2023). When large traders create noise. *Journal of Financial Economics*, 150(2), 103709.
- Grossman, S. (1976). On the efficiency of competitive stock markets where trades have diverse information. *The Journal of Finance*, 31(2), 573–585.
- Grossman, S. J., and Stiglitz, J. E. (1980). On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3), 393–408.
- Hellwig, M. F. (1980). On the aggregation of information in competitive markets. *Journal of Economic Theory*, 22(3), 477–498.
- Hellwig, M. F. (1982). Rational expectations equilibrium with conditioning on past prices: A mean-variance example. *Journal of Economic Theory*, 26(2), 279–312.
- Holden, C. W., and Subrahmanyam, A. (1992). Long-lived private information and imperfect competition. *The Journal of Finance*, 47(1), 247–270.
- Jackson, M. O. (1991). Equilibrium, price formation, and the value of private information. *The Review of Financial Studies*, 4(1), 1–16.
- Jackson, M. O. (2003). Efficiency and information aggregation in auctions with costly information. *Review of Economic Design*, 8(2), 121–141.
- Kacperczyk, M., Nosal, J., and Sundaresan, S. (2025). Market power and price informativeness. *Review of Economic Studies*, 92(3), 1955–1986.
- Kacperczyk, M., Sundaresan, S., and Wang, T. (2021). Do foreign institutional investors improve price efficiency? *The Review of Financial Studies*, 34(3), 1317–1367.
- Kothari, S. P., and Sloan, R. G. (1992). Information in prices about future earnings: Implications for earnings response coefficients. *Journal of Accounting and Economics*, 15(2-3), 143–171.
- Kovalenkov, A., and Vives, X. (2014). Competitive rational expectations equilibria without apology. *Journal of Economic Theory*, 149, 211–235.

- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, 1315–1335.
- Kyle, A. S. (1989). Informed speculation with imperfect competition. *The Review of Economic Studies*, 56(3), 317–355.
- Lambert, N. S., Ostrovsky, M., and Panov, M. (2018). Strategic trading in informationally complex environments. *Econometrica*, 86(4), 1119–1157.
- Lee, J., and Kyle, A. S. (2018). When are financial markets perfectly competitive?
- Lee, S. (2013). *Active investment, liquidity externalities, and markets for information* [Available at SSRN 1300706].
- Martinelli, C. (2006). Would rational voters acquire costly information? *Journal of Economic Theory*, 129(1), 225–251.
- Milgrom, P. R. (1981). Rational expectations, information acquisition, and competitive bidding. *Econometrica: Journal of the Econometric Society*, 921–943.
- Parlour, C. A., and Rajan, U. (2003). Payment for order flow. *Journal of Financial Economics*, 68(3), 379–411.
- Parlour, C. A., and Seppi, D. J. (2003). Liquidity-based competition for order flow. *The Review of Financial Studies*, 16(2), 301–343.
- Peress, J. (2010). The tradeoff between risk sharing and information production in financial markets. *Journal of Economic Theory*, 145(1), 124–155.
- Pesendorfer, W., and Swinkels, J. M. (2000). Efficiency and information aggregation in auctions. *American Economic Review*, 90(3), 499–525.
- Subrahmanyam, A. (1991). Risk aversion, market liquidity, and price efficiency. *The Review of Financial Studies*, 4(3), 417–441.
- Verrecchia, R. E. (1982). Information acquisition in a noisy rational expectations economy. *Econometrica: Journal of the Econometric Society*, 1415–1430.
- Vives, X. (2002). Private information, strategic behavior, and efficiency in cournot markets. *RAND Journal of Economics*, 361–376.
- Vives, X. (2011). Strategic supply function competition with private information. *Econometrica*, 79(6), 1919–1966.
- Vives, X. (2014). On the possibility of informationally efficient markets. *Journal of the European Economic Association*, 12(5), 1200–1239.
- Zhu, H. (2014). Do dark pools harm price discovery? *The Review of Financial Studies*, 27(3), 747–789.

A Proofs and derivations

A.1 Trading equilibrium (Proof of Lemma 1)

The existence and uniqueness of a linear equilibrium is a direct consequence of Theorem 1 in Lambert et al. (2018).

Because throughout this proof $\mathbf{z}$ is fixed, we call, with a slight abuse of notation $\mathbf{p} = \mathbf{p}(\cdot, \mathbf{z})$, and, for each trader i , $\mathbf{d}_i = \mathbf{d}_i(\cdot, \mathbf{z})$ —that is, we drop $\mathbf{z}$ from the arguments of these functions.

To characterize the linear equilibrium, notice that, under the equilibrium strategies, the pricing function from the point of view of trader i is:

$$\mathbf{p}(Y) = \lambda Y = \lambda d_i + \lambda \sum_{j \neq i} \beta_j s_j + \lambda u = \lambda d_i + \lambda \sum_{j \neq i} \beta_j v + \lambda \sum_{j \neq i} \beta_j \varepsilon_j + \lambda u.$$

Taking that into account, the first-order-condition for trader i determines:

$$d_i = \frac{1}{2\lambda} \left(1 - \lambda \sum_{j \neq i} \beta_j \right) \mathbb{E}[v|s_i] = \frac{1}{2\lambda} \left(1 - \lambda \sum_{j \neq i} \beta_j \right) \frac{\tau_i}{\tau_v + \tau_i} s_i,$$

where the second equality follows from Bayesian updating given joint normality of (v, s_i) and $\varepsilon_i \perp v$. Recall that in a linear equilibrium, $d_i = \beta_i s_i$. Define $B = \sum_j \beta_j$, and $\kappa_i = \frac{\tau_i}{\tau_i + \tau_v}$. The equation above is then equivalent to:

$$\beta_i = \frac{1}{2\lambda} (1 - \lambda(B - \beta_i)) \kappa_i, \tag{A.1}$$

which allows us to solve for β_i as a function of λ and B :

$$\beta_i = \underbrace{\frac{\kappa_i}{2 - \kappa_i}}_{z_i} \frac{1 - \lambda B}{\lambda}.$$

By summing both sides up and solving for B :

$$B = \frac{\mathbf{z} \cdot \mathbf{1}}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{1}{\lambda}, \tag{A.2}$$

and plugging into the expression of β_i :

$$\beta_i = \frac{z_i}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{1}{\lambda}. \tag{A.3}$$

We now obtain an expression for λ by using the market-maker's breakeven condition.

To that goal, define the unbiased signal from aggregate demand:

$$\tilde{Y} = \frac{Y}{B} = v + \underbrace{\frac{\sum_j \beta_j \varepsilon_j + u}{B}}_{\eta},$$

with $\eta \perp v$ normally distributed with precision

$$\tau_\eta := \text{Var}[\eta]^{-1} = \frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}}.$$

Then, by standard Bayesian updating for joint normal variables:

$$\mathbb{E}[v|Y] = \mathbb{E}[v|\tilde{Y}] = \frac{\frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}}}{\frac{B^2}{\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1}} + \tau_v} \tilde{Y} = \frac{B}{B^2 + \tau_v \left(\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1} \right)} Y,$$

where the first equality holds because Y and $\tilde{Y}$ contain the same information.

Because $\mathbb{E}[v|Y] = p = \lambda Y$, we obtain an expression for λ :

$$\lambda = \frac{B}{B^2 + \tau_v \left(\sum_j \beta_j^2 \tau_j^{-1} + \tau_u^{-1} \right)}.$$

Towards a closed-form solution for λ , we will obtain simplifications for parts of the expression above. First, using $\tau_i^{-1} = \frac{1}{2\tau_v} \frac{1-z_i}{z_i}$, and equation (A.3):

$$\sum_j \beta_j^2 \tau_j^{-1} = \frac{1}{2} \frac{\mathbf{z} \cdot \mathbf{1} - \mathbf{z} \cdot \mathbf{z}}{(1 + \mathbf{z} \cdot \mathbf{1})^2 \lambda^2} \tau_v^{-1}.$$

Plugging the equation above and equation (A.2) in the expression for λ , one can solve for λ to obtain:

$$\frac{1}{L(\mathbf{z})} = \lambda = \sqrt{\frac{\tau_u}{2\tau_v}} \frac{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}}{1 + \mathbf{z} \cdot \mathbf{1}},$$

which established our result on liquidity, and $\beta_i = \sqrt{\frac{2\tau_v}{\tau_u}} \frac{z_i}{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}}$.

We now characterize price informativeness. Because prices contain the same information as Y , which contains the same information as $\tilde{Y}$, $\text{Var}[v|p] = \text{Var}[v|\tilde{Y}]$. By standard Bayesian updating for jointly normal variables:

$$\text{Var}[v|\tilde{Y}]^{-1} = \tau_v + \tau_\eta = \tau_v(1 + \mathbf{z} \cdot \mathbf{1}).$$

But then:

$$I(\mathbf{z}) = 1 - \frac{\text{Var}[v|\tilde{Y}]}{\text{Var}[v]} = \frac{\mathbf{z} \cdot \mathbf{1}}{1 + \mathbf{z} \cdot \mathbf{1}},$$

as we wanted to prove.

All that is left is to calculate the ex-ante utility. For that, note that $\mathbb{E}[p|s_i] = \lambda \mathbb{E}[Y|s_i] = \lambda \sum_{j \neq i} \beta_j \kappa_i s_i + \lambda \beta_i s_i$. Moreover, by equation (A.1), $2\lambda \beta_i = (1 - \lambda \sum_{j \neq i} \beta_j) \kappa_i$. Thus:

$$\mathbb{E}[\mathbf{d}_i(s_i)(v - \mathbf{p}(Y))|s_i] = \beta_i s_i^2 (2\lambda \beta_i - \lambda \beta_i) = \lambda \beta_i^2 s_i^2.$$

Taking ex-ante utility and recalling that, ex-ante, s_i has mean zero, we have $\mathbb{E}[s_i^2] = \text{Var}[s_i] = \tau_v^{-1} + \tau_i^{-1}$, which delivers, after algebra:

$$V_i(z_i, \mathbf{z}) = \lambda \beta_i^2 (\tau_v^{-1} + \tau_i^{-1}) = \sqrt{\frac{1}{2\tau_v \tau_u}} \frac{1}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{z_i + z_i^2}{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}},$$

which can be rewritten as in the statement of the proposition. This concludes the proof. $\blacksquare$

A.2 Symmetric equilibrium (Statement and Proof of Lemma A.1)

In this section we characterize the symmetric information equilibrium for any finite n , and consider their asymptotic limits.

Lemma A.1. *Let z^n denote a symmetric equilibrium information acquisition in the economy with n strategic traders, in which $p^n = \lambda^n Y$ and $d^n = \beta^n s_i$ represent prices and (symmetric) demand on the equilibrium path. Then,*

$$\begin{aligned} \lambda^n &= \sqrt{\frac{\tau_u}{2\tau_v}} \sqrt{\frac{nz^n(1+z^n)}{(1+nz^n)^2}} \\ Y^n &= \sqrt{\frac{2\tau_v}{\tau_u}} \sqrt{\frac{nz^n}{1+z^n}} \left(\frac{1}{n} \sum_{i=1}^n s_i \right) + \tau_u^{-1/2} e_u \\ p^n &= \frac{nz^n}{1+nz^n} \left(\frac{1}{n} \sum_{i=1}^n s_i \right) + \sqrt{\frac{\tau_v^{-1}}{2}} \frac{\sqrt{nz^n(1+z^n)}}{1+nz^n} e_u, \end{aligned}$$

where $e_u = u / \tau_u^{-1/2} \sim \text{Normal}(0,1)$. Assume $Z^* := \lim_{n \rightarrow \infty} nz^n \in [0, \infty]$, the large- n limit of

aggregate information, exists. Then, we have that

$$\frac{1}{n} \sum_{i=1}^n s_i \rightarrow v + \sqrt{\frac{\tau_v^{-1}}{2Z^*}} e_s,$$

in distribution, where $e_s \sim \text{Normal}(0,1)$ is independent of v and u . Consequently, the large- n limiting equilibrium objects are, in distribution,

$$\begin{aligned} \lim_{n \rightarrow \infty} \lambda^n &= \sqrt{\frac{\tau_v^{-1}}{2\tau_u^{-1}}} \frac{\sqrt{Z^*}}{1 + Z^*} \\ \lim_{n \rightarrow \infty} Y^n &= \sqrt{2\tau_u^{-1}} \left[\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right] \\ \lim_{n \rightarrow \infty} p^n &= \tau_v^{-1/2} \frac{\sqrt{Z^*}}{1 + Z^*} \left[\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right]. \end{aligned}$$

In [Appendix C.1](#), we also prove an analogous version of [Lemma A.1](#) for an asymmetric equilibrium where each trader chooses to be fully-informed with probability π_n^* , which is the type of equilibrium that arises with fixed information costs. The results are identical.

A.2.1 Proof of [Lemma A.1](#)

In a symmetric equilibrium, $\mathbf{z} = z^n \mathbf{1}$, and so $\mathbf{z} \cdot \mathbf{1} = nz^n$ and $\mathbf{z} \cdot \mathbf{z} = n(z^n)^2$. Substituting these results into the expressions obtained in the proof of [Lemma 1](#) for λ^n and β^n

$$\begin{aligned} \lambda^n &= \sqrt{\frac{\tau_u}{2\tau_v}} \sqrt{\frac{nz^n(1 + z^n)}{(1 + nz^n)^2}}, \\ \beta^n &= \frac{1}{n} \sqrt{\frac{2\tau_v}{\tau_u}} \sqrt{\frac{nz^n}{1 + z^n}}. \end{aligned}$$

Then, noting $e_u := u / \tau_u^{-1/2} \sim \text{Normal}(0,1)$, we have

$$\begin{aligned} Y^n &= \sqrt{\frac{2\tau_v}{\tau_u}} \sqrt{\frac{nz^n}{1 + z^n}} \frac{\sum_i s_i}{n} + \tau_u^{-1/2} e_u, \\ p^n &= \frac{nz^n}{1 + nz^n} \frac{\sum_i s_i}{n} + \sqrt{\frac{\tau_v^{-1}}{2}} \frac{\sqrt{nz^n(1 + z^n)}}{1 + nz^n} e_u. \end{aligned}$$

We will now take the limit $n \rightarrow \infty$. First, note the fact that $s = v + \varepsilon_i$ in a symmetric

equilibrium, where

$$\varepsilon \sim \mathcal{N}\left(0, \frac{1}{2}\tau_v^{-1} \frac{1-z^n}{z^n}\right).$$

Therefore,

$$\frac{\sum_i \varepsilon_i}{n} \sim \text{Normal}\left(0, \frac{1}{2}\tau_v^{-1} \frac{1-z^n}{nz^n}\right)$$

and so using the definition $Z^* := \lim_{n \rightarrow \infty} nz^n$ and the result that $z^n \rightarrow 0$ (Lemma 2), we have by the Central Limit Theorem,

$$\frac{\sum_i s_i}{n} \rightarrow v + \sqrt{\frac{\tau_v^{-1}}{2Z^*}} e_s$$

in distribution, where $e_s \sim \text{Normal}(0,1)$. Consequently, regardless of whether Z^* is finite or infinite, we have the following limits in distribution,

$$\begin{aligned} Y_n &\rightarrow \sqrt{2\tau_u^{-1}} \left(\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right) \\ p^n &\rightarrow \tau_v^{-1/2} \frac{\sqrt{Z^*}}{1+Z^*} \left(\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right) \end{aligned}$$

This proves Lemma A.1. ■

A.3 Vanishing individual information (Proof of Lemma 2)

First, consider the case $\chi = \infty$. Given strict convexity of c and the fact that $\frac{dV_i(z \mathbf{1})}{dz_i} < \infty$ for all $z > 0$, the only possible solution to equation (5) is $z^n = 0$.

Now, suppose $\chi < \infty$, so that $z^n > 0$ for each n . Consider a convergent sub-sequence $(n_t)_{t \geq 0}$ such that $\lim_{t \rightarrow \infty} z^{n_t} = z^* > 0$. In that case, we have $\lim_{t \rightarrow \infty} n_t z^{n_t} = +\infty$. Thus, we have $\lim_{t \rightarrow \infty} \frac{dV_i(z^{n_t} \mathbf{1})}{dz_i} = 0$. By equation (5), we must also have $\lim_{t \rightarrow \infty} c'(z^{n_t}) = 0$. Since c is strictly convex, we have by the monotone convergence theorem $0 = \lim_{t \rightarrow \infty} c'(z^{n_t}) = c'(z^*)$. But strict convexity also implies $c'(z) > 0$ for all $z > 0$, contradicting $z^* > 0$. This argument works for any sub-sequence, and so $\lim_{n \rightarrow \infty} z^n$ exists and is equal to zero. ■

A.4 Aggregate information characterization (Proof of Theorem 1)

The case of $\chi = \infty$ (and therefore $\Gamma = 0$) is already proved in Lemma 2, which showed that $z^n = 0$ for each n .

Next, we prove the case $\chi = 0$ (and therefore $\Gamma = \infty$). By Lemma 2, we know that z^n is a positive sequence that converges to zero. Assume a bounded subsequence of z^n exists. Consider a subsequence such that $n_t z^{n_t} \rightarrow Z < \infty$ —the existence of which is a consequence of the existence of a bounded subsequence. Then, we have $\lim_{t \rightarrow \infty} \frac{dV_i(z^{n_t} \mathbf{1})}{dz_i} > 0$. On the other hand, the fact that $z^{n_t} \rightarrow 0$ implies by the monotone convergence theorem that $c'(z^{n_t}) \rightarrow \chi = 0$. But, along that subsequence, we have that (5) holds with equality, which contradicts the fact that $\frac{dV_i(z^{n_t} \mathbf{1})}{dz_i}$ and $c'(z^{n_t})$ have different limit points. Therefore, each subsequence of z_n must be unbounded, and $n z^n \rightarrow \infty$, which proves that $\mathcal{I} = 1$.

Finally, we prove the claim for $\chi \in (0, \infty)$ —and so $\Gamma \in (0, \infty)$. By Lemma 2, we have $c'(z^n) \rightarrow \chi > 0$, which implies $\frac{dV_i(z^{n_t} \mathbf{1})}{dz_i} \rightarrow \chi$ for each subsequence $(n_t)_{t \geq 0}$, by equation (5). If $z^{n_t} \rightarrow 0$ and $n_t z^{n_t} \rightarrow \infty$, $\frac{dV_i(z^{n_t} \mathbf{1})}{dz_i} \rightarrow 0$, which contradicts the equality, which means that each subsequence is bounded, and therefore the sequence $n z^n$ is itself bounded. Take any converging subsequence $(n_t)_{t \geq 0}$ and define $Z := \lim_{t \rightarrow \infty} n_t z^{n_t}$. We then obtain:

$$\begin{aligned} \chi &= \lim_{t \rightarrow \infty} \frac{dV_i(z^{n_t} \mathbf{1})}{dz_i} \\ &= \lim_{t \rightarrow \infty} \sqrt{\frac{1}{2\tau_v \tau_u}} \frac{1}{1 + n_t z^{n_t}} \frac{1}{\sqrt{n_t z^{n_t} + n_t (z^{n_t})^2}} \\ &= \sqrt{\frac{1}{2\tau_v \tau_u}} \frac{1}{1 + Z} \frac{1}{\sqrt{Z}}, \end{aligned}$$

which can be reorganized to obtain: $Z(1 + Z)^2 = \Gamma$. Because there is a unique positive root Z , for this equation, call it Z^* , we conclude that all converging subsequences converge to Z^* . Because $n z^n$ was shown to be bounded, in the first place, then $n z^n \rightarrow Z^*$. To conclude, it suffices to show $Z^* = \frac{\mathcal{I}}{1 - \mathcal{I}}$, so that we obtain the equality in the statement of the theorem.

To that end, recall from Lemma A.1 that limiting prices satisfy:

$$p := \lim_{n \rightarrow \infty} p^n = \tau_v^{-1/2} \frac{\sqrt{Z^*}}{1 + Z^*} \left[\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right].$$

Define $\tilde{p}$ as the signal extracted from the price that is unbiased for v . That is, $\tilde{p} = p \frac{1 + Z^*}{Z^*} = v + \zeta$, where $\zeta \perp v$ is normally distributed with precision: $\tau_\zeta = Z^* \tau_v$. Therefore,

$$\text{Var}[v|\tilde{p}] = \text{Var}[v|p] = \frac{1}{Z^* \tau_v + \tau_v'},$$

and therefore

$$\mathcal{I} = 1 - \frac{\text{Var}[v|p]}{\text{Var}[v]} = \frac{Z^*}{1 + Z^*}.$$

Reorganizing this equality, we obtain $Z^* = \frac{\mathcal{I}}{1-\mathcal{I}}$, proving the theorem. $\blacksquare$

A.5 Market observables (Proofs of Proposition 1 and Proposition 3)

The results follow immediately from the limiting characterization of prices in Lemma A.1, the limiting characterization of informativeness in Theorem 1, and the result that the limit $\mathcal{I}$ exists (shown in the proof of Theorem 1). $\blacksquare$

A.6 Growing noise case (Proof of Proposition 2)

We know the following first-order condition has to hold in any symmetric information equilibrium:

$$\frac{dV_i(z^n, z^n \mathbf{1})}{dz_i} \leq c'(z^n).$$

Starting from the value function

$$V_i(z_i, \mathbf{z}) = \sqrt{\frac{\tau_v^{-1} \tau_{u,n}^{-1}}{2}} \frac{1}{1 + \mathbf{z} \cdot \mathbf{1}} \frac{z_i + z_i^2}{\sqrt{\mathbf{z} \cdot \mathbf{1} + \mathbf{z} \cdot \mathbf{z}}},$$

calculated in Lemma 1, we obtain its derivative when evaluated at the symmetric z^n :

$$\begin{aligned} \frac{dV_i(z^n, z^n \mathbf{1})}{dz_i} &= \frac{V_i(z^n, z^n \mathbf{1})}{z^n} \left(\frac{1 + 2z^n}{1 + z^n} - \frac{z^n}{1 + nz^n} - \frac{z^n + 2(z^n)^2}{nz^n + n(z^n)^2} \right) \\ &= \frac{V_i(z^n, z^n \mathbf{1})}{z^n} \left(\frac{n-1}{n} \frac{1 + 2z^n}{1 + z^n} - \frac{z^n}{1 + nz^n} \right). \end{aligned}$$

Notice that all the accumulation points of the sequence in parentheses are in a bounded strictly positive set. Indeed, because

$$0 \leq \frac{z^n}{1 + nz^n} \leq \frac{1}{n},$$

we must have:

$$\begin{aligned} 1 &\leq \liminf \frac{1+2z^n}{1+z^n} = \liminf \left(\frac{n-1}{n} \frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n} \right) \\ &\leq \limsup \left(\frac{n-1}{n} \frac{1+2z^n}{1+z^n} - \frac{z^n}{1+nz^n} \right) = \limsup \frac{1+2z^n}{1+z^n} \leq 2. \end{aligned}$$

As a consequence, the limiting behavior of $\frac{dV_i(z^n, z^n \mathbf{1})}{dz_i}$ follows from the limiting behavior of $\frac{V_i(z^n, z^n \mathbf{1})}{z^n}$. Henceforth, we want to study:

$$\frac{V_i(z^n, z^n \mathbf{1})}{z^n} = \sqrt{\frac{\tau_v^{-1} \tau_{u,n}^{-1}}{2}} \frac{1}{1+nz^n} \frac{\sqrt{1+z^n}}{\sqrt{nz^n}}.$$

First, note that for any large but finite n we must have $z^n > 0$. Indeed, if $z^n = 0$, the term above is infinity, which violates our first-order condition. Therefore, the first-order condition must hold with equality for sufficiently large n .

We consider the case of $\chi > 0$ (the case $\chi = 0$ follows easily since aggregation information grows even faster in that case). From the first-order condition and the results above, we then have $\liminf \frac{V_i(z^n, z^n \mathbf{1})}{z^n} = k \liminf c'(z^n)$, where $k \geq \frac{1}{2}$. Since $c'(z^n) \geq \chi$ for all n (since c is convex), we therefore have, asymptotically:

$$\liminf \sqrt{\tau_{u,n}^{-1}} = K \liminf ((nz^n)^{1/2} + (nz^n)^{3/2})$$

where $K \geq k\chi \frac{\sqrt{2\tau_v}}{\limsup \sqrt{1+z^n}}$ is a positive constant.

If $\tau_{u,n}^{-1}$ grows unbounded, the dominant term in the right-hand side is the one with the higher exponent and so:

$$nz^n \sim \tau_{u,n}^{-1/3}.$$

Because nz^n explodes, it immediately follows that $\mathcal{I} = 1$ and $\mathcal{L} = \infty$.

Regarding the existence of a free-entry limit in this growing-noise case, suppose that $\limsup n^{-3} \tau_{u,n}^{-1} = 0$ as assumed in the proposition. Then, from $nz^n \sim \tau_{u,n}^{-1/3}$, we have that $z^n \rightarrow 0$. At the same time, $nz^n \sim \tau_{u,n}^{-1/3}$ also implies $\frac{V_i(z^n, z^n \mathbf{1})}{z^n} \sim \bar{v}$ for some positive constant $\bar{v}$. Combining these two results, $V_i(z^n, z^n \mathbf{1}) \rightarrow 0$. Since we also have $c(z^n) \rightarrow c(0) = 0$, this shows that traders make zero profits, i.e., $\lim V_i(z^n, z^n \mathbf{1}) - c(z^n) = 0$. Thus, a free-entry equilibrium exists. This concludes our proof.

While not required by the proposition, we also consider the case if $\tau_{u,n}^{-1}$ decreases to zero. In that case, the dominant term in the right-hand-side is the one with the lower

exponent and so:

$$nz^n \sim \tau_{u,n}^{-1}.$$

In that case, it immediately follows that $\mathcal{I} = 0$ while $\mathcal{L}$ remains bounded. ■

A.7 Identifying Price Informativeness

We now provide a proof-of-concept on how to identify informativeness and the primitives of our model from observables. To that end, we leverage the closed-form solutions obtained in [Theorem 1](#), [Proposition 1](#), and [Proposition 3](#). Although there are several methodologies suitable for identifying informativeness (Bai et al., 2016; Dávila and Parlato, 2023, 2025; Farboodi et al., 2022; Kacperczyk et al., 2021; Kothari and Sloan, 1992), our approach overcomes a number of difficulties faced by previous work. Concretely, to identify informativeness, the existing literature needs to make assumptions about which fundamentals prices are supposed to predict, and at which time window. Moreover, because the usual empirical methods rely on accounting data, price-informativeness can only be identified at low-frequencies. Our structural approach sidesteps these issues entirely, at the cost of taking the model seriously: it uses equilibrium variables as observables, and relies on the model mapping to identify both the primitives of the model as well as price informativeness.

Disentangling changes in fundamental and non-fundamental uncertainty is essential to measure changes in informativeness. In what follows, we will demonstrate how to decompose variation in widely available data into these underlying, unobservable shocks. To formalize this approach, we consider an econometrician who can observe a dataset consisting of time series of trading volume $\mathcal{Y}$ and price volatility $\mathcal{V}$. The goal of the econometrician is to estimate, from the shifts in the two available variables, shocks to fundamental and non-fundamental volatility, as well as changes in price-informativeness—i.e., variation in τ_v^{-1} , τ_u^{-1} and $\mathcal{I}$, respectively.

Note first that price volatility, trading volume, and liquidity are tightly connected by the shape of the pricing function. Indeed, the expressions in [Proposition 1](#) and [Proposition 3](#) imply

$$\mathcal{L} = \frac{\mathcal{Y}}{\mathcal{V}^2}.$$

Given data on volatility and volume, we can use our model to infer liquidity, even though it may be hard to measure directly.

Going forward, it is thus equivalent to assume that the dataset contains observations

of liquidity and volatility, rather than volume and volatility. Formally, we will define the set of observables to be $\{\mathcal{L}_t, \mathcal{V}_t\}_{t \in \{0,1,\dots,T\}}$. Our results will rely on two identifying assumptions, imposed in sequence. The first assumption allows us to describe exactly the evolution of the variables of interest as a function of the evolution of observables, up to the level of aggregate precision in the market, which is unobservable.

Assumption 1. *Information costs— χ —do not change throughout the dataset.*

Assumption 1 allows us to attribute any changes in observables to variation in fundamental and non-fundamental noise. It states that while beliefs about fundamentals and non-fundamental demand may change at a high frequency, the same does not hold for information costs, which tend to move only slowly.

Proposition A.1. *Under **Assumption 1**, the evolution of informativeness, fundamental precision, and non-fundamental precision follows:*

$$d \log \mathcal{I} = \frac{1 - \mathcal{I}}{1 + \mathcal{I}} (d \log \mathcal{L} + 2d \log \mathcal{V}), \quad (\text{A.4})$$

$$d \log \tau_v = \frac{1 - \mathcal{I}}{1 + \mathcal{I}} d \log \mathcal{L} - \frac{4\mathcal{I}}{1 + \mathcal{I}} d \log \mathcal{V}, \quad (\text{A.5})$$

$$d \log \tau_u = -\frac{2 + \mathcal{I}}{1 + \mathcal{I}} d \log \mathcal{L} - \frac{2}{1 + \mathcal{I}} d \log \mathcal{V}. \quad (\text{A.6})$$

Given an initial value for informativeness, $\mathcal{I}_0$ and given observables $\mathcal{L}_0, \mathcal{V}_0$, one can obtain the initial values for $\tau_{v,0}$ and $\tau_{u,0}$.

Proposition A.1 provides a method to calculate precisely how price informativeness and uncertainty vary using observable variables. Given the observable dataset $\{\mathcal{L}_t, \mathcal{V}_t\}$, the only additional object one needs to implement **Proposition A.1** is the initial level of informativeness $\mathcal{I}_0$. While $\mathcal{I}_0$ may be hard to observe or estimate directly, one can perform sensitivity analyses: sample the input $\mathcal{I}_0$ from a statistical prior, compute paths for $(\mathcal{I}_t, \tau_{v,t}, \tau_{u,t})_{t \geq 0}$, and then ask whether the range of paths (one path for each $\mathcal{I}_0$) shares common properties. Of course, the outcome depends on the data for $\{\mathcal{L}_t, \mathcal{V}_t\}$ as well as the choice of prior for $\mathcal{I}_0$, but the process is easily implementable with **Proposition A.1**.

An alternative approach is to rely on the reduced-form estimates in the literature to choose $\mathcal{I}_0$. Remarkably, estimates of relative price informativeness tend to be low. Dávila and Parlato (2025), for example, obtain values close to 2.5%. Our next identifying assumption allows us to approximate the dynamics of the market under the assumption that informativeness is low.

Assumption 2. *Informativeness is low: $\mathcal{I} \approx 0$*

Corollary 3. Under *Assumption 1* and *Assumption 2*, the evolution of informativeness, fundamental uncertainty, and non-fundamental noise can be approximated as follows:

$$d\log \mathcal{I} = d\log \mathcal{L} + 2d\log \mathcal{V} + o(\mathcal{I}), \quad (\text{A.7})$$

$$d\log \tau_v = d\log \mathcal{L} + o(\mathcal{I}), \quad (\text{A.8})$$

$$d\log \tau_u = -2(d\log \mathcal{L} + d\log \mathcal{V}) + o(\mathcal{I}). \quad (\text{A.9})$$

These expressions allow for immediate implementation of the empirical strategy on readily available data. It is instructive to compare our results with Dávila and Parlato (2025). That paper develops a strategy to identify the level of price informativeness by running a regression of current prices on future asset payoffs, a valid approach in a large class of models—including ours—when primitives are assumed to be constant. Our approach differs from theirs in two ways. First, we are interested in identifying time-variation in price informativeness, which arises exactly from changes in primitives: our key assumption is that movements in informativeness follow from high-frequency shifts in fundamental uncertainty and non-fundamental noise. Second, instead of using the level of prices and asset payoffs, we use two observable quantities—volatility and volume—to disentangle the effects of each shock on informativeness.

A.7.1 Proof of Proposition A.1

We start with the observable metrics for $\mathcal{V}$ and $\mathcal{L}$ from Proposition 1-Proposition 3, and the equation for informativeness in Theorem 1. Taking logs on both side of these equations, recalling χ is assumed to be fixed and using total differentiation we obtain:

$$\begin{aligned} d\log \mathcal{V} &= \frac{1}{2}(d\log \mathcal{I} - d\log \tau_v), \\ d\log \mathcal{L} &= \frac{1}{2}(d\log \tau_v - d\log \tau_u - \log \mathcal{I} - \log(1 - \mathcal{I})), \\ d\log \mathcal{I} - 3d\log(1 - \mathcal{I}) &= -d\log \tau_u - d\log \tau_v. \end{aligned}$$

By the chain rule, we rewrite $d\log(1 - \mathcal{I}) = -\frac{\mathcal{I}}{1-\mathcal{I}}d\log \mathcal{I}$. Substituting that in the equations above, and solving for $d\log \tau_v, d\log \tau_u, d\log \mathcal{I}$ as a function of $d\log \mathcal{V}$ and $d\log \mathcal{L}$, we conclude:

$$d\log \mathcal{I} = \frac{1 - \mathcal{I}}{1 + \mathcal{I}} (d\log \mathcal{L} + 2d\log \mathcal{V}), \quad (\text{A.10})$$

$$d\log \tau_v = \frac{1 - \mathcal{I}}{1 + \mathcal{I}} d\log \mathcal{L} - \frac{4\mathcal{I}}{1 + \mathcal{I}} d\log \mathcal{V}, \quad (\text{A.11})$$

$$d\log \tau_u = -\frac{2 + \mathcal{I}}{1 + \mathcal{I}} d\log \mathcal{L} - \frac{2}{1 + \mathcal{I}} d\log \mathcal{V}. \quad (\text{A.12})$$

Given an initial value for $\mathcal{I}_0$, and initial observables $\mathcal{L}_0, \mathcal{V}_0$, we can use the characterization of equilibrium to calculate $\tau_{v,0}$ and $\tau_{u,0}$. So given $\mathcal{I}_0$ we can obtain the complete dynamics of the system. ■

B A Model of Limit Orders and Risk Aversion

We maintain the assumption that there is one risky asset with payoff $v \sim \mathcal{N}(0, \tau_v^{-1})$, noise trader demand is $u \sim \mathcal{N}(0, \tau_u^{-1})$, and traders observe signal $s_i = v + \varepsilon_i$, with $\varepsilon_i \sim \mathcal{N}(0, \tau_i^{-1})$. Moreover, $u \perp v \perp \varepsilon_i$ for all i , and $\varepsilon_i \perp \varepsilon_j$, for all i, j . Trader i 's expected payoff of buying d units of the asset at price p is:

$$u_i(x) = \frac{1 - \mathbb{E} \left[e^{-\gamma(v-p)d} \right]}{\gamma}.$$

Besides the payoff difference, there are two other changes relative to the baseline model. First, the trader submits a demand schedule, specifying for each possible price p the amount $d_i(p)$ of the asset he demands, taking into account market-clearing. Second, there is a competitive fringe—also understood as free entry—of uninformed traders. This assumption is often used in the literature (Bernhardt and Taub, 2006; Vives, 2014), and it leads to a simplification of the model: market clearing is substituted by the unbiasedness of prices. Formally, given aggregate demand $Y = \sum_i d_i + u$, for all informed traders i :

$$p = \mathbb{E}[v|Y].$$

B.1 Proof of Theorem 2

We start by assuming a linear equilibrium for a fixed number of informed entrants, n . That is, in this equilibrium each trader submits demand

$$d_i = B_i s_i + C_i Y \quad (\text{B.1})$$

And prices satisfy

$$p = \lambda Y \quad (\text{B.2})$$

We start by noticing that the definition of Y and the aggregation of individual demand leads to:

$$Y = \sum_j B_j s_j + \underbrace{C}_{\equiv \sum_j C_j} Y + u, \iff Y = \frac{1}{1 - C} \left(\sum_j B_j s_j + u \right).$$

Notice that in this model $\frac{\partial p}{\partial d_i} = \lambda \frac{1}{1 - \sum_{j \neq i} C_j}$, for any i . By the mean-variance representation of the individual problem, we must have:

$$d_i = \frac{\mathbb{E}[v|p, s_i] - p}{\gamma \text{Var}[v|p, s_i] + \lambda \frac{1}{1 - \sum_{j \neq i} C_j}} \quad (\text{B.3})$$

We will now obtain the Gaussian filtering. For that, define, for any player i , the auxiliary variable W_i which orthogonalizes the aggregate order flow and their signal:

$$W_i = \frac{(1 - C)Y - B_i s_i}{\underbrace{B - B_i}_{\equiv \sum_{j \neq i} B_j}} = v + \frac{\sum_j B_j \varepsilon_j + u}{\underbrace{B - B_i}_{\equiv \eta_i}}. \quad (\text{B.4})$$

We then have that $\eta_i \perp (v, \varepsilon_i)$, and

$$\tau_{\eta_i} \equiv \text{Var}[\eta_i]^{-1} = \frac{(B - B_i)^2}{\sum_{j \neq i} B_j^2 \tau_j^{-1} + \tau_u^{-1}}.$$

This reformulation allows us to think of the posterior precision of v as linear in the precisions of the orthogonal signals. That is:

$$\kappa_i \equiv \text{Var}[v|p, s_i]^{-1} = \tau_v + \tau_i + \tau_{\eta_i}, \quad (\text{B.5})$$

and

$$\mathbb{E}[v|p, s_i] = \frac{\tau_i}{\kappa_i} s_i + \frac{\tau_{\eta_i}}{\kappa_i} W_i = \frac{\tau_i (B - B_i) - \tau_{\eta_i} B_i}{\kappa_i (B - B_i)} s_i + \frac{\tau_{\eta_i} (1 - C)}{\kappa_i (B - B_i)} Y. \quad (\text{B.6})$$

Then, we can plug (B.5) and (B.6) in the mean-variance expression for the demand of agent i , equation (B.3), and equate that to the linear assumption in (B.1). Matching coefficients in those expressions we get:

$$B_i \left(\lambda \frac{1}{1 - \sum_{j \neq i} C_j} \kappa_i + \gamma \right) + \frac{B_i}{B - B_i} \tau_{\eta_i} = \tau_i, \quad (\text{B.7})$$

and

$$C_i \left(\lambda \frac{1}{1 - \sum_{j \neq i} C_j} \kappa_i + \gamma \right) = \frac{1 - C}{B - B_i} \tau_{\eta_i} - \lambda \kappa_i \quad (\text{B.8})$$

Note that because τ_{η_i} depends only on B 's and κ_i depends only on τ_{η_i} , the system of equations in (B.7) solves for B 's as a function of λ . With that information, one can then solve (B.8) as a function of λ . So we just need to close the equilibrium by learning λ . That is obtained by recalling that $p = \mathbb{E}[v|Y]$.

We start by making Y unbiased for v . That is, define:

$$\tilde{Y} \equiv \frac{1 - C}{B} Y = v + \underbrace{\frac{\sum_j B_j \varepsilon_j + u}{B}}_{\eta}.$$

We have that $\tau_\eta = \underbrace{\sum_j \frac{B_j^2}{B_j^2 \tau_j^{-1} + \tau_u^{-1}}}_{\equiv D}$. Thus:

$$\lambda Y \equiv p = \frac{B^2/D}{\tau_v + \frac{B^2}{D}} \frac{(1 - C)}{B} Y \iff \lambda = \frac{B(1 - C)}{\tau_v D + B^2}. \quad (\text{B.9})$$

Finally, we can specialize to the case of interest, in which trader 1 has precision τ_1 and all other traders have precision τ^* and are, therefore, symmetric. In that case, the equilibrium equations become:

$$B_1 \left(\lambda \frac{1}{1 - (n - 1)C^*} \kappa_1 + \gamma \right) + \frac{B_1}{(n - 1)B^*} \tau_{\eta_1} = \tau_1, \quad (\text{B.10})$$

and

$$B^* \left(\lambda \frac{1}{1 - (n - 2)C^* - C_1} \kappa^* + \gamma \right) + \frac{B^*}{(n - 2)B^* + B_1} \tau_{\eta^*} = \tau^*, \quad (\text{B.11})$$

with $\kappa_1 = \tau_v + \tau_1 + \tau_{\eta_1}$, $\kappa^* = \tau_v + \tau^* + \tau_{\eta^*}$, and with

$$\tau_{\eta_1} = \frac{[(n-1)B^*]^2}{\frac{[(n-1)B^*]^2}{(n-1)\tau^*} + \frac{1}{\tau_u}}, \quad \tau_{\eta^*} = \frac{\left(\frac{n-2}{n-1}(n-1)B^* + B_1\right)^2}{\frac{B_1^2}{\tau_1} + \frac{n-2}{n-1} \frac{[(n-1)B^*]^2}{(n-1)\tau^*} + \frac{1}{\tau_u}} \quad (\text{B.12})$$

and $\lambda = \frac{B(1-C)}{\tau_v D + B^2}$, with $D = \frac{B_1^2}{\tau_1} + \frac{[(n-1)B^*]^2}{(n-1)\tau^*} + \frac{1}{\tau_u}$. We also have equations for C , but those will play a minor role in what follows.

We now calculate the indirect payoff of trader 1 in a linear equilibrium. Before submitting a demand schedule, the trader observes his signal s_i . Therefore, his demand optimization must solve:

$$\begin{aligned} d_i(p) &\in \arg \min_x \mathbb{E}_v \left[e^{-\gamma(v-p)x} | (p, s_i) \right] \\ &= \arg \max_x \mathbb{E}[(v-p)x | p, s_i] - \frac{\gamma}{2} x^2 \text{Var}[v | p, s_i] \end{aligned}$$

The first-order-condition to this problem satisfies:

$$d_i(p) = \frac{\mathbb{E}[v - p | p, s_i]}{\gamma \text{Var}[v | p, s_i] + \frac{dp}{dd_i}}.$$

In a linear equilibrium, it will be the case that $\frac{dp}{dd_i} = \gamma \lambda_i$ for some constant. Therefore, the ex-ante indirect utility of trader i is:

$$\begin{aligned} \gamma V_i &:= 1 - \mathbb{E} \left[\mathbb{E}_v [e^{-\gamma(v-p)d_i(p)} | p, s_i] \right] = 1 - \mathbb{E} \left[e^{-\gamma \left(\mathbb{E}[(v-p)d_i(p) | p, s_i] - \frac{\gamma}{2} x_i^2(p) \text{Var}[v | p, s_i] \right)} \right] \\ &= 1 - \mathbb{E} \left[e^{-\gamma^2 \frac{1}{2} (\text{Var}[v | p, s_i] + 2\lambda_i) d_i(p)^2} \right], \end{aligned}$$

where the last observation follows from the first-order-condition for $d_i(p)$. Under a linear equilibrium, $d_i(p)$ will be an ex-ante mean zero gaussian random variable. Because $\text{Var}[v | p, s_i]$ is a constant across realizations of p and s_i , the term in the exponent is a constant multiplying a normal random variable. Therefore, up to normalizing $d_i(p)$, the expectation is the m.g.f. of a chi-squared distribution, and therefore:

$$\gamma V_i = 1 - \left(1 + \gamma^2 (\text{Var}[v | p, s_i] + 2 \lambda_i) \text{Var}[d_i(p)] \right)^{-1/2}.$$

Finally, recall that

$$d_i(p) = \frac{\mathbb{E}[v - p | p, s_i]}{\gamma (\text{Var}[v | p, s_i] + \lambda_i)},$$

and therefore

$$\text{Var}[d_i(p)] = \frac{\text{Var}[\mathbb{E}[v - p|p, s_i]]}{\gamma^2(\text{Var}[v|p, s_i] + \lambda_i)^2} = \frac{\text{Var}[v - p] - \text{Var}[v|p, s_i]}{\gamma^2(\text{Var}[v|p, s_i] + \lambda_i)^2},$$

where the second equality uses the law of total variance and the fact that $\text{Var}[v|p, s_i]$ is constant in realizations of (p, s_i) . We thus conclude that:

$$V_i = 1 - \left(\frac{\text{Var}[v - p] + \frac{\text{Var}[v - p] + \lambda_i}{\text{Var}[v|p, s_i] + \lambda_i} \lambda_i}{\text{Var}[v|p, s_i] + \lambda_i} \right)^{-1/2}.$$

Noting that $\lambda_i = \frac{1}{\gamma} \lambda \frac{1}{1 - \sum_{j \neq i} C_j}$:

$$V_i = \frac{1}{\gamma} \left(1 - \left(\frac{\text{Var}[v - p] + \frac{\text{Var}[v - p] + \lambda \frac{1}{\gamma} \frac{1}{1 - (n-1)C^*}}{\text{Var}[v|p, s_i] + \lambda \frac{1}{\gamma} \frac{1}{1 - (n-1)C^*}} \lambda \frac{1}{\gamma} \frac{1}{1 - (n-1)C^*}}{\text{Var}[v|p, s_i] + \lambda \frac{1}{\gamma} \frac{1}{1 - (n-1)C^*}} \right)^{-1/2} \right).$$

We now use our price equation in the semi-symmetric equilibrium:

$$p - v = \lambda Y - v = \frac{\lambda}{1 - C} \left[Bv + \sum_j B_j \varepsilon_j + u \right] - v = \frac{(\lambda B - (1 - C))}{1 - C} v + \frac{\lambda}{1 - C} B \eta$$

Thus,

$$\text{Var}[p - v] = \left[\frac{(\lambda B - (1 - C))}{1 - C} \right]^2 \tau_v^{-1} + \left[\frac{\lambda}{1 - C} B \right]^2 \frac{D}{B^2}.$$

By substituting the formula for λ from (B.9) above, we obtain:

$$\kappa_p \equiv \text{Var}[v - p]^{-1} = \tau_v + \frac{B^2}{D}.$$

Moreover, we know $\text{Var}[v|p, s_1] = \kappa_1^{-1}$. By defining $R \equiv \lambda \frac{1}{\gamma} \frac{1}{1 - (n-1)C^*}$, we obtain:

$$V_1(\tau_1, \tau^* \mathbf{1}) = \frac{1}{\gamma} \left(1 - \left(\frac{\kappa_p + \kappa_1 \kappa_p R}{\kappa_1 + \frac{\kappa_1 + \kappa_1 \kappa_p R}{\kappa_p + \kappa_1 \kappa_p R} \kappa_1 \kappa_p R} \right)^{1/2} \right)$$

By unbiasedness of prices, it must be the case that $V_1 = 0$ when $\tau_1 = 0$. We want to find the derivative of V_1 with respect to τ_1 , for any fixed level of aggregate information

$T \equiv (n-1)\tau^*$, when n is large. That derivative is the highest constant ζ_T such that:

$$\underbrace{\frac{1}{2\gamma} \left(1 - (1 - \gamma V_1(\tau_1, \tau^* \mathbf{1}))^2 \right)}_{\approx V_1'(0, \tau^* \mathbf{1}) \tau_1 \text{ for small } \tau_1} = \frac{1}{2\gamma} \frac{\kappa_1 - \kappa_p}{\kappa_1} \frac{1 + \frac{\kappa_1 R}{1 + \kappa_1 R}}{1 + \kappa_1 R \frac{1 + \kappa_p R}{1 + \kappa_1 R}} \geq \zeta_T \tau_1 + O(\tau_1^2),$$

Notice that $\frac{B^2}{D} = \tau_\eta$ is the informativeness of the aggregate order flow. So κ_p is the precision of v conditional on the price.

Notice that:

$$\kappa_1 - \kappa_p = \tau_1 + (\tau_{\eta_i} - \tau_\eta) = \tau_1 + \left(\frac{[(n-1)B^*]^2}{\frac{[(n-1)B^*]^2}{T} + \frac{1}{\tau_u}} - \frac{((n-1)B^* + B_1)^2}{\frac{B_1^2}{\tau_1} + \frac{[(n-1)B^*]^2}{T} + \frac{1}{\tau_u}} \right) \quad (\text{B.13})$$

By equations (B.11) and (B.10), we can obtain upper bounds for B^* and B_1 . Indeed:

$$(n-1)B^* \leq \frac{T}{\gamma},$$

and

$$B_1 \leq \frac{\tau_1}{\gamma}.$$

Define the variables $\bar{B} = (n-1)B^*$ and $\bar{C} = (n-1)C^*$.

Recall that $\tau_\eta = \frac{(\bar{B} + B_1)^2}{\frac{B_1^2}{\tau_1} + \frac{\bar{B}^2}{T} + \frac{1}{\tau_u}}$. The inequality above shows that, for small τ_1 , $\tau_\eta \approx \frac{\bar{B}^2}{\frac{\bar{B}^2}{T} + \frac{1}{\tau_u}}$.

Notice, at the same time, that for small τ_1 , $\tau_{\eta_1} \approx \tau_\eta$.

Moreover, for large enough n , one can easily see that $\tau_{\eta^*} \approx \tau_\eta$. We now use (B.8) to prove $C_1 \approx 0$ and, $\bar{C} \approx C$ when n is large. We have

$$C_1 \left(\frac{B(1-C)}{\tau_v D + B^2} \frac{1}{1 - \bar{C}} \kappa_1 + \gamma \right) = \frac{1-C}{\bar{B}} \tau_{\eta_i} - \frac{B(1-C)}{\tau_v D + B^2} \kappa_1 \quad (\text{B.14})$$

and

$$C^* \left(\frac{B(1-C)}{\tau_v D + B^2} \frac{1}{1 - \frac{(n-2)}{(n-1)} \bar{C} - C_1} \kappa^* + \gamma \right) = \frac{1-C}{\frac{n-2}{n-1} \bar{B} + B_1} \tau_{\eta^*} - \frac{B(1-C)}{\tau_v D + B^2} \kappa^* \quad (\text{B.15})$$

Because when τ_1 is small, $\tau_\eta \approx \frac{\bar{B}^2}{\frac{\bar{B}^2}{T} + \frac{1}{\tau_u}} = \frac{B^2}{D}$, we have $\tau_{\eta_i} \approx \tau_\eta$, and $\kappa_i \approx \tau_v + \frac{B^2}{D} = \frac{\tau_v D + B^2}{D}$.

Plugging these in (B.14)

$$C_1 \left(\frac{B(1-C)}{\tau_v D + \bar{B}^2} \frac{1}{1-\bar{C}} \kappa_1 + \gamma \right) \approx \frac{(1-C)\bar{B}}{D} - \frac{\bar{B}(1-C)}{D} = 0,$$

and therefore $C_1 \approx 0$, and $\bar{C} \approx C$. Thus, when τ_1 is small and n is large, we have from (B.11):

$$\bar{B} \left(\lambda \frac{1}{1-C} \kappa_p + \gamma + \frac{1}{\bar{B}} \tau_\eta \right) \approx T,$$

and from (B.10):

$$B_1 \left(\lambda \frac{1}{1-C} \kappa_p + \gamma + \frac{1}{\bar{B}} \tau_\eta \right) \approx \tau_1.$$

Therefore, B_1 is approximately linear with τ_1 for large n and low τ_1 . That is:

$$B_1 \approx \frac{\bar{B}}{T} \tau_1.$$

We can then use (B.16) to write:

$$\kappa_1 - \kappa_p \approx \tau_1 + \left(\frac{\bar{B}^2}{\frac{\bar{B}^2}{T} + \frac{1}{\tau_u}} - \frac{(\bar{B} + \frac{\bar{B}}{T} \tau_1)^2}{\frac{\bar{B}^2}{T^2} \tau_1 + \frac{\bar{B}^2}{T} + \frac{1}{\tau_u}} \right) \quad (\text{B.16})$$

By rewriting this equation, we obtain, for small τ_1 :

$$\kappa_1 - \kappa_p \approx \left(1 + \frac{\tau_\eta}{T} \left(\frac{\tau_\eta}{T} - 2 \right) \right) \tau_1 = \left(1 - \frac{\tau_\eta}{T} \right)^2 \tau_1.$$

Moreover, for fixed T , large n and τ_1 close to zero we know that $\kappa_1 = \kappa_p = \tau_v + \frac{\bar{B}^2}{D}$. We also know

$$R\kappa_p = \frac{\lambda}{\gamma} \frac{1}{1-C} \left(\tau_v + \frac{\bar{B}^2}{D} \right) = \frac{1}{\gamma} \frac{\bar{B}}{\tau_v D + \bar{B}^2} \left(\frac{\tau_v D + \bar{B}^2}{D} \right) = \frac{1}{\gamma} \frac{\bar{B}}{D}.$$

Therefore, the derivative of trader 1's value at $\tau_1 = 0$ is:

$$V'_1(0, \tau^* \mathbf{1}) = \frac{1}{2\gamma} \frac{1}{\tau_v + \tau_\eta} \frac{1 + \frac{2}{\gamma} \frac{\tau_\eta}{\bar{B}}}{1 + \frac{2}{\gamma} \frac{\tau_\eta}{\bar{B}} + [\frac{1}{\gamma} \frac{\tau_\eta}{\bar{B}}]^2} \left(1 - \frac{\tau_\eta}{T} \right)^2, \quad \text{where} \quad \tau_\eta = \frac{B^2}{\frac{\bar{B}^2}{T} + \frac{1}{\tau_u}}.$$

Equilibrium is defined by a pair of equations: $V'_1(0, \tau^* \mathbf{1}) = \frac{\lambda}{2\tau_v}$, and, by summing all

the matching coefficient equations for B :¹²

$$T \approx \bar{B} \left(\lambda \frac{1}{1-C} \kappa_p + \gamma + \frac{1}{\bar{B}} \tau_\eta \right) = B \left(2 \frac{B}{D} + \gamma \right).$$

Instead of solving for B , which is the aggregate demand-signal sensitivity, and T , which is the total amount of precision, we use the change-of-variables

$$Z := \frac{\frac{B^2}{\tau_v}}{\frac{B^2}{T} + \frac{1}{\tau_u}} \quad \text{and} \quad \varphi := \frac{\frac{B^2}{T}}{\frac{B^2}{T} + \frac{1}{\tau_u}}.$$

It is useful to observe that Z is $\frac{\tau_\eta}{\tau_v}$, that is, Z is the precision of the signal in the aggregate demand—and thus in the price—about v , normalized by the precision of v . We then get $\mathcal{I} = \frac{Z}{1+Z}$. Moreover, $\varphi = \frac{\tau_\eta}{T}$ is the percentage of the aggregate market precision that leaks through prices.

With these variables, we have the relationships $T = \tau_v \frac{Z}{\varphi}$ and $B^2 = \frac{Z}{1-\varphi} \tau_v$. Plugging this into the cubic equation for B , we recover

$$\begin{aligned} 1 - 2\varphi &= \frac{\gamma\varphi}{\sqrt{1-\varphi}} \frac{1}{\sqrt{Z\tau_v\tau_u}} \\ \Rightarrow (1 - 2\varphi) \frac{\sqrt{1-\varphi}}{\sqrt{\varphi}} &= \frac{\gamma}{\sqrt{T\tau_u}} \end{aligned} \tag{B.17}$$

Plugging the change-of-variables into the information FOC $U'_1(0) = c'(0) = \frac{\chi}{2\tau_v}$, we get

$$\frac{1}{2\gamma\tau_v} \frac{1-2\varphi}{1+Z} = \frac{\chi}{2\tau_v}$$

or written differently

$$\frac{1}{2\tau_v} \frac{1}{\sqrt{Z}(1+Z)} \frac{\varphi}{\sqrt{1-\varphi}} \frac{1}{\sqrt{\tau_v\tau_u}} = \frac{\chi}{2\tau_v}$$

We want to keep this form in hand to develop intuition later (because the left-hand-side is exactly $U'_1(0)$). However, we also want to simplify this equation to obtain a cleaner

¹²The adjustment for χ comes from observing that V_1 was written as a function of τ_1 and not z_1 . Noticing that $\frac{dz_1}{d\tau_1}(0) = \frac{1}{2\tau_v}$ gives us the adjustment.

expression. After some algebra, we obtain the expression

$$Z(1+Z)^2 = \Gamma \frac{2\varphi^2}{1-\varphi}, \quad \text{where} \quad \Gamma := \frac{1}{2\chi^2\tau_v\tau_u} \quad (\text{B.18})$$

From these two equations, we can easily take the limit $\gamma \rightarrow 0$ to get

$$\begin{aligned} \lim_{\gamma \rightarrow 0} \varphi &= \frac{1}{2} \\ (\lim_{\gamma \rightarrow 0} Z)(1 + \lim_{\gamma \rightarrow 0} Z)^2 &= \Gamma \end{aligned}$$

These equations are identical to the market order model. But in general since $\varphi \leq \frac{1}{2}$, strictly so when $\gamma > 0$, equation (B.18) shows that $Z(1+Z)^2 \leq \frac{1}{2}\Gamma$, meaning that Z is always smaller in this model than the baseline model. Using the fact that $\mathcal{I} = \frac{Z}{1+Z}$, we can also manipulate equations (B.17) and (B.18) to rewrite them as

$$\begin{aligned} \varphi &= \frac{1 - \frac{\gamma\chi}{1-\mathcal{I}}}{2} \\ \frac{\mathcal{I}}{(1-\mathcal{I})^2} \frac{1-\varphi}{2\varphi^2} &= \Gamma \end{aligned}$$

exactly as the theorem requires. From the first equation, we clearly see the requirement for equilibrium is $\gamma\chi \leq 1$, as stated in the theorem.

It is useful to rewrite these two equations as one. Plugging (B.17) into (B.18), we need only to solve a single nonlinear equation for Z .

$$Z(1+Z)^2 = \frac{1}{2}\Gamma \frac{(1 - \gamma\chi(1+Z))^2}{1 + \gamma\chi(1+Z)} \quad (\text{B.19})$$

Because the left-hand-side is increasing in Z and the right-hand-side is decreasing in Z , this equation has at most one solution. Setting $Z = 0$, the left-hand-side is lower than the right-hand-side (or equal if $\Gamma = 0$). Setting $Z = \frac{1}{\gamma\chi} - 1$, we have that the left-hand-side is larger if these parameters are such that Z is positive. Thus a positive solution exists as long as $\gamma\chi < 1$.

To show the comparative statics, rewrite (B.19) as:

$$Z(1+Z)^2 \frac{1 + \gamma\chi(1+Z)}{(1 - \gamma\chi(1+Z))^2} = \frac{1}{2}\Gamma,$$

The right-hand-side is decreasing in τ_u , and the left-hand-side is increasing in Z . Thus,

Z decreases in τ_u , and so does $\mathcal{I}$. Although not required for the theorem, note that the same holds for τ_v , since it enters Γ symmetrically to τ_u .

Finally, the expression from liquidity follows from $\frac{dp}{du} = \frac{\lambda}{1-C}$. ■

B.2 Comparative statics (Statement and Proof of **Proposition B.1**)

Proposition B.1. *The following comparative statics hold:*

- Price volatility increases in fundamental and noise volatility—i.e., $\frac{d\mathcal{V}}{d\tau_v^{-1}}, \frac{d\mathcal{V}}{d\tau_u^{-1}} > 0$;
- Liquidity decreases in fundamental and increases in noise volatility—i.e., $\frac{d\mathcal{L}}{d\tau_v^{-1}} < 0 < \frac{d\mathcal{L}}{d\tau_u^{-1}}$.

We shall show the result using the more general model in this section, which allows for risk aversion. As previously established, the fact that the model above has the investor submit limit orders is immaterial.

B.2.1 Proof of **Proposition B.1**

Volatility. Because free-entry prices satisfy $p = \mathbb{E}[v|Y]$, we have the following identity:

$$\begin{aligned} \text{Var}[p] &= \text{Var}[\mathbb{E}[v|Y]] = \text{Var}[v] - \mathbb{E}[\text{Var}[v|Y]] = \text{Var}[v] - \mathbb{E}[\text{Var}[v|p]] \\ &= \text{Var}[v] - \text{Var}[v|p] = \tau_v^{-1} \mathcal{I}, \end{aligned}$$

where the second equality holds by the Law of total variance, the third equality follows from $p = \lambda Y$, the fourth equality from $\text{Var}[v|p]$ being independent of the realization of p by joint normality, and the last equality from the definition of $\mathcal{I}$. Therefore, $\mathcal{V} = \sqrt{\tau_v^{-1} \mathcal{I}}$.

In the proof of **Theorem 2**, we have shown informativeness is increasing with τ_u^{-1} and τ_v^{-1} (the latter is not stated directly, but the argument is identical to the one for τ_u , since the statistic Γ is symmetric in the two precisions). Given these comparative statics for $\mathcal{I}$, the comparative statics of $\mathcal{V}$ follow directly.

Liquidity. Rewriting equation (B.19) as a function of $\mathcal{I} = \frac{Z}{1+Z}$, we obtain the following:

$$\frac{\mathcal{I}}{(1-\mathcal{I})^3} = \frac{1}{2} \frac{1}{\chi^2 \tau_u \tau_v} \frac{(1 - \frac{\gamma\chi}{1-\mathcal{I}})^2}{1 + \frac{\gamma\chi}{1-\mathcal{I}}}. \quad (\text{B.20})$$

Note that the final ratio can be written:

$$\frac{(1 - \frac{\gamma\chi}{1-\mathcal{I}})^2}{1 + \frac{\gamma\chi}{1-\mathcal{I}}} = \frac{(1 - \gamma\chi - \mathcal{I})^2}{(1-\mathcal{I})(1 + \gamma\chi - \mathcal{I})}.$$

Plugging the above in equation (B.20) and solving for τ_v :

$$\tau_v = \frac{1}{2\chi^2\tau_u} \frac{(1-\mathcal{I})^2(1-\gamma\chi-\mathcal{I})^2}{\mathcal{I}(1+\gamma\chi-\mathcal{I})}. \quad (\text{B.21})$$

Therefore, using the expressions for liquidity and φ in Theorem 2:

$$\mathcal{L} = \sqrt{\frac{2\tau_u^{-1}}{\tau_v^{-1}}} \frac{1}{\sqrt{\mathcal{I}(1-\mathcal{I})(1+\frac{\gamma\chi}{1-\mathcal{I}})}} = \frac{\tau_u^{-1}}{\chi} \frac{(1-\mathcal{I})(1-\gamma\chi-\mathcal{I})}{\mathcal{I}(1+\gamma\chi-\mathcal{I})}. \quad (\text{B.22})$$

Finally, notice that $\mathcal{L}$ is decreasing as a function of $\mathcal{I}$, all else equal. Indeed, as $\mathcal{I}$ increases, $\frac{1-\mathcal{I}}{\mathcal{I}}$ decreases. Moreover, we see that

$$\frac{1-\gamma\chi-\mathcal{I}}{1+\gamma\chi-\mathcal{I}} = 1 - 2\gamma\chi \frac{1}{1+\gamma\chi-\mathcal{I}}$$

is also decreasing in $\mathcal{I}$.

We are now ready to prove the result for τ_v . First, notice that, in (B.22), $\mathcal{L}$ does not depend on τ_v directly. If τ_v^{-1} increases, we have proved $\mathcal{I}$ increases, and thus $\mathcal{L}$ decreases, since we proved $\mathcal{L}$ is decreasing in $\mathcal{I}$ for fixed τ_u^{-1} .

For the τ_u^{-1} result, we start again from (B.20). By the expression for $\mathcal{L}$ above, equation (B.20) can be rewritten as, after taking square root:

$$2\chi\tau_v = \mathcal{L}(1-\mathcal{I})(1-\gamma\chi-\mathcal{I}).$$

Notice that the equality above does not depend on τ_u^{-1} directly. If τ_u^{-1} grows, $\mathcal{I}$ increases, so the terms depending on $\mathcal{I}$ decrease. Therefore, to maintain equality, $\mathcal{L}$ has to increase. ■

C A model with fixed costs of information

Now, let's assume the information cost satisfies $\bar{\chi} := c(z) > 0$ for all z . There is a fixed cost of acquiring any information. The fact that there is no additional variable cost simplifies the analysis so that either $z_i = 0$ or $z_i = 1$ for each trader i (none or perfect information).

One concern with fixed costs, suggested by the analysis of Grossman and Stiglitz (1980), is that an equilibrium may fail to exist. We will show that this is not the case here. In fact, there is a well-defined equilibrium for any $\bar{\chi}$. First, we will consider

symmetric equilibria, followed by asymmetric equilibria. The key statistic, which is exactly analogous to (9), is

$$\Gamma := \frac{1}{2\bar{\chi}^2\tau_u\tau_v}. \quad (\text{C.1})$$

Whether Γ is greater or less than 2 determines the type of equilibrium. And the level of Γ determines the level of aggregate information in a large market.

In a symmetric equilibrium, the pre-information-cost utility value of each trader, assuming all other traders collect information z^n , is the same as before:

$$V_n(z; z^n) = \sqrt{\frac{\tau_v^{-1}\tau_u^{-1}}{2}} \frac{z(1+z)}{(1+(n-1)z^n+z)\sqrt{(n-1)z^n(1+z^n)+z(1+z)}}.$$

Notice that $V_n(0;0) = V_n(0;1) = 0$. The key criterion for equilibrium is optimality: in an equilibrium with information collection, trader utility must satisfy

$$V_n(1;1) - V_n(0;1) \geq \bar{\chi}, \quad \text{if } z^n = 1. \quad (\text{C.2})$$

On the other hand, for an equilibrium with ignorance, trader utility must satisfy

$$V_n(1;0) - V_n(0;0) \leq \bar{\chi}, \quad \text{if } z^n = 0. \quad (\text{C.3})$$

An asymmetric equilibrium is more complicated. To set it up, let π_n^* denote the equilibrium probability a trader acquires any information. Because each trader only acquires either $z_i = 0$ or $z_i = 1$, aggregate information in the market will be equal to the total number of informed traders, which is the binomial random variable $\text{Binom}(n, \pi_n^*)$. The pre-cost value of a single trader, assuming the fraction of the other $n-1$ traders acquiring information is π_n , is

$$V_n(z; \pi_n) = \sqrt{\frac{\tau_v^{-1}\tau_u^{-1}}{2}} \mathbb{E} \left[\frac{z(1+z)}{(1+(n-1)\pi_n+z)\sqrt{2(n-1)\pi_n+z(1+z)}} \right],$$

where the expectation is over the possible realizations of π_n . By rational expectations, this random variable has probability distribution $(n-1)\pi_n \sim \text{Binom}(n-1, \pi_n^*)$. In equilibria of the large- n economy, the amount of randomness in π_n vanishes, i.e., $\pi_n - \pi_n^* \rightarrow 0$ almost-surely, by the law of large numbers. In a mixed-strategy equilibrium with $\pi_n^* > 0$

being the information probability, it must be the case that

$$V_n(1; \pi_n) - \bar{\chi} = V_n(0; \pi_n) = 0, \quad \text{if } \pi_n \in (0, 1),$$

i.e., the trader is indifferent between acquiring information or not.

Based on this characterization, we can prove the following theorem, which is roughly speaking the fixed cost version of [Theorem 1](#).

Theorem C.1. *An equilibrium with fixed information costs always exists in the large- n limit and satisfies the following:*

1. If $\Gamma \leq 2$, then the equilibrium is symmetric and features $nz^n = 0$ for all n large enough.
2. If $\Gamma > 2$, then the equilibrium is asymmetric and features $n\pi_n^* \rightarrow \Pi^*$, where Π^* is the unique positive solution to $(\Pi + 2)^2(\Pi + 1) = \Gamma^2$.

Proof. 1. Suppose $\Gamma \leq 2$. First, we prove that, assuming a symmetric equilibrium exists, $nz^n = 0$ for all n large enough. This is a direct consequence of $\lim_{n \rightarrow \infty} V_n(1; 1) = 0$, implying $V_n(1) < \bar{\chi}$ for all n large enough, so (C.2) cannot hold. Second, we prove existence of a symmetric equilibrium with $z^n = 0$ for large n . The utility from deviating from $z^n = 0$ is $V(1; 0) = \frac{1}{2} \sqrt{\tau_v^{-1} \tau_u^{-1}}$. Thus, $V_n(1; 0) - V_n(0; 0) - \bar{\chi} = \frac{1}{2} \sqrt{\tau_v^{-1} \tau_u^{-1}} - \bar{\chi} \leq 0$, so (C.3) holds, and the equilibrium is confirmed.

2. Suppose $\Gamma > 2$. As a consequence of the argument for claim 1, a symmetric equilibrium cannot occur if $\Gamma > 2$, because in such case neither (C.2) nor (C.3) can hold. Moving to asymmetric equilibria, it is easy to see that $n\pi_n^* \not\rightarrow \infty$ as $n \rightarrow \infty$. Indeed, if $n\pi_n^* \rightarrow \infty$ occurred, then $V_n(1; \pi_n^*) \rightarrow 0$, and so $V_n(1; \pi_n) \rightarrow 0$ a.s. (recall $\pi_n - \pi_n^* \rightarrow 0$ by the law of large numbers). In fact, writing $\Pi^* := \lim_{n \rightarrow \infty} n\pi_n^*$ and then evaluating $V_n(1; \pi_n) = \bar{\chi}$ in the large- n limit, we find that Π^* must solve the cubic equation

$$(\Pi + 2)^2(\Pi + 1) = \frac{\tau_v^{-1} \tau_u^{-1}}{\bar{\chi}^2}.$$

This has a unique positive solution $\Pi^* > 0$ if and only if $\Gamma > 2$. Thus, an asymmetric equilibrium exists if $\Gamma > 2$, but not if $\Gamma \leq 2$, and the aggregate information in this equilibrium is Π^* as $n \rightarrow \infty$. $\square$

C.1 Asymmetric equilibrium

In this section, we prove a result analogous to [Lemma A.1](#) but for the fixed information cost case covered above in [Theorem C.1](#) (with $\Gamma > 2$). In such case, an “asymmetric

equilibrium" arises where each trader chooses to be fully-informed with probability π_n^* .

Lemma C.1. *Let π_n^* denote the probability of information acquisition in an asymmetric equilibrium with n strategic traders, and let $\Pi_n := \text{Binom}(n, \pi_n^*)$ be the aggregate number of informed traders. Then,*

$$\begin{aligned}\beta_n &= \sqrt{\frac{\tau_v^{-1}}{\tau_u^{-1}} \frac{\Pi_n}{(1 + \Pi_n)^2}} \\ Y_n &= \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}} \Pi_n v + \tau_u^{-1/2} e_u \\ p^n &= \tau_v^{-1/2} \left(\frac{\Pi_n}{1 + \Pi_n} \frac{v}{\tau_v^{-1/2}} + \frac{\sqrt{\Pi_n}}{1 + \Pi_n} e_u \right)\end{aligned}$$

where $e_u = u / \tau_u^{-1/2} \sim \text{Normal}(0, 1)$. Let $\Pi^* := \lim_{n \rightarrow \infty} n \pi_n^* \in [0, \infty]$ be the large- n limit of aggregate information. Then, we have the following limiting equilibrium objects, almost-surely,

$$\begin{aligned}\lim_{n \rightarrow \infty} \beta_n &= \sqrt{\frac{\tau_v^{-1}}{\tau_u^{-1}} \frac{\sqrt{\Pi^*}}{1 + \Pi^*}} \\ \lim_{n \rightarrow \infty} Y_n &= \tau_u^{-1/2} \left[\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right] \\ \lim_{n \rightarrow \infty} p^n &= \tau_v^{-1/2} \frac{\sqrt{\Pi^*}}{1 + \Pi^*} \left[\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right]\end{aligned}$$

Proposition C.1. *In the large- n limit of asymmetric equilibria, the measures of liquidity, price informativeness, excess price volatility, and trading volume are given by the following:*

$$\begin{aligned}(\text{liquidity}) \quad \mathcal{L} &= \sqrt{\frac{\tau_v}{\tau_u} \frac{1 + \Pi^*}{\sqrt{\Pi^*}}} \\ (\text{informativeness}) \quad \mathcal{I} &= \frac{\Pi^*}{1 + \Pi^*} \\ (\text{volatility}) \quad \mathcal{V} &= \tau_v^{-1} \frac{\Pi^*}{(1 + \Pi^*)} \\ (\text{volume}) \quad \mathcal{Y} &= \sqrt{\tau_u^{-1} \tau_v^{-1} \Pi^*}\end{aligned}$$

Proof. Suppose in the size- n economy, each trader obtains a perfect signal with probability π_n^* , and otherwise remains uninformed. In this case, $\mathbf{z}$ is a vector of zeros and ones, with the probability of each non-zero entry being an independent Bernoulli draw. Then, $\mathbf{z} \cdot \mathbf{1} = \|\mathbf{z}\|^2 := \Pi_n \sim \text{Binom}(n, \pi_n^*)$. Substituting this into formulas in the proof of 1, we

have

$$\lambda^n = \sqrt{\frac{\tau_v^{-1}}{\tau_u^{-1}} \frac{\Pi_n}{(1 + \Pi_n)^2}} \quad \text{and} \quad \beta^n s = \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}} \Pi_n v,$$

and so

$$\begin{aligned} Y_n &= \sqrt{\frac{\tau_u^{-1}}{\tau_v^{-1}}} \Pi_n v + \tau_u^{-1/2} e_u \\ p^n &= \tau_v^{-1/2} \left(\frac{\Pi_n}{1 + \Pi_n} \frac{v}{\tau_v^{-1/2}} + \frac{\sqrt{\Pi_n}}{1 + \Pi_n} e_u \right) \end{aligned}$$

To take the limit $n \rightarrow \infty$, note that $\Pi_n / (n\pi_n^*)$ converges to 1 almost-surely, by the Strong Law of Large Numbers. Hence, denoting $\Pi^* := \lim_{n \rightarrow \infty} n\pi_n^*$, we have $\Pi_n \rightarrow \Pi^*$ almost-surely. Using this fact, we have

$$\begin{aligned} Y_n &\rightarrow \tau_u^{-1/2} \left(\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right) \\ p^n &\rightarrow \tau_v^{-1/2} \frac{\sqrt{\Pi^*}}{1 + \Pi^*} \left(\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right) \end{aligned}$$

This proves [Lemma C.1](#).

[Proposition C.1](#) is proved by combining the results of [Lemma C.1](#) with the definitions of the measures. $\square$

Comparing the result above with [3](#), we see that the limiting objects are very similar across the symmetric and asymmetric equilibria (which recall arise with variable and fixed costs, respectively). In particular, modulo some $\sqrt{2}$ constants, the formulas are identical in distribution. For instance, compare the limiting price in the two equilibrium types:

$$\begin{aligned} \text{(symmetric equilibrium)} \quad \lim_{n \rightarrow \infty} p^n &= \tau_v^{-1/2} \frac{\sqrt{Z^*}}{1 + Z^*} \left[\sqrt{Z^*} \frac{v}{\tau_v^{-1/2}} + \frac{e_s + e_u}{\sqrt{2}} \right] \\ \text{(asymmetric equilibrium)} \quad \lim_{n \rightarrow \infty} p^n &= \tau_v^{-1/2} \frac{\sqrt{\Pi^*}}{1 + \Pi^*} \left[\sqrt{\Pi^*} \frac{v}{\tau_v^{-1/2}} + e_u \right] \end{aligned}$$

Since Z^* and Π^* have similar interpretations, as aggregate information (in units of precision), the formulas are almost identical. The only difference is that the noise $\frac{e_s + e_u}{\sqrt{2}}$ in the symmetric equilibrium price is replaced by e_u in the asymmetric equilibrium price. But since both are standard normal random variables, the prices are identical in distribution.

For this reason, the measures in [Proposition 3](#) and [Proposition C.1](#), for symmetric and asymmetric equilibria, are identical.

[Theorem C.1](#) proves that equilibrium exists generically. In competitive models like Grossman and Stiglitz (1980), equilibrium non-existence is a problem: neither a fully-informed nor a fully-uninformed equilibrium can exist. But here, we can obtain either of these extreme cases depending on the size of the statistic Γ . If $\Gamma \leq 2$, we have a fully-uninformed equilibrium. But if $\Gamma \rightarrow \infty$, we have a fully-informed equilibrium, in the sense that aggregate information becomes maximal. The characterization of the result in terms of $\Pi^* = \lim_{n \rightarrow \infty} n\pi_n^*$ is intuitive: Π^* represents the *number of informed traders* in the large- n limit, which necessarily remains finite for any fixed cost $\bar{\chi}$.

D Comparison to Competitive NREE

We compare our results to a competitive Noisy Rational Expectations Equilibrium (NREE) setting along the lines of Grossman (1976), Grossman and Stiglitz (1980), Hellwig (1980), and Verrecchia (1982). Because we will ignore strategic considerations, we now need traders to be risk averse. We assume trader i 's utility function is of the CARA type:

$$\tilde{V}_i := -\exp(-\gamma[d_i(v - p) - c(z_i)]),$$

where z_i is the signal precision acquired, and $c(\cdot)$ its cost, analogous to the baseline model. In the trading stage, traders receive their signal $s_i = v + \varepsilon_i$ and solve

$$\max_{d_i} \mathbb{E}[\tilde{V}_i \mid s_i, p]$$

In the information stage, traders choose their precision z_i by solving

$$\max_{z_i} \mathbb{E}\left[\max_{d_i} \mathbb{E}[\tilde{V}_i \mid s_i, p]\right]$$

As in the baseline model, we assume signal errors ε_i are independent across traders and independent of both the fundamental v and the noise u . Furthermore, recall that the precision z_i is defined to be related to the overall signal variance $\sigma_i := \text{Var}[s_i]$ in that

$$\sigma_i = \sigma(z_i) := \frac{1}{2}\tau_v^{-1}(z_i^{-1} + 1)$$

To keep the proper comparison to the baseline model, traders pay a cost to increase z_i , but the results would be very similar if the cost was defined over the precision of the

signal error $(\sigma_i - \tau_v^{-1})^{-1} = \frac{2}{\tau_v^{-1}} \frac{z_i}{1-z_i}$.¹³ The equilibrium asset price is determined via the market clearing condition

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n d_i + u = 0$$

It is standard in the NREE literature to clear markets based on the *average* demand, by analogy to the continuum limit, rather than the total demand; equivalently, some versions explicitly model the noise u grow with the number of traders (i.e., u is the noise per capita).¹⁴ Finally, conjecture (and later verify) that the equilibrium pricing function takes the form

$$p = \beta_v v + \beta_u u,$$

for some β_v and β_u .

In this setup, joint normality of (v, u, s_i) implies the distribution of (s_i, p) is

$$(s_i, p) \sim \text{Normal}(0, \tilde{\Sigma}_i),$$

where $\tilde{\Sigma}_i := \begin{pmatrix} \sigma_i & \beta_v \tau_v^{-1} \\ \beta_v \tau_v^{-1} & \beta_v^2 \tau_v^{-1} + \beta_u^2 \tau_u^{-1} \end{pmatrix}$

¹³ Note that Verrecchia (1982) defines a cost function $\tilde{c}(s)$ over the error precision $s(z) := \frac{2}{\tau_v^{-1}} \frac{z}{1-z}$. We assume that this cost of precision is convex, i.e., $\tilde{c}''(s) \geq 0$. Given our representation of the cost function as $c(z)$, this is implemented simply by putting $\tilde{c}(s) := c(z^{-1}(s)) = c(\frac{\tau_v^{-1}s}{2+\tau_v^{-1}s})$. Note that for $\tilde{c}''(s)$ to be convex, we require that $(1-z)^2 c'(z)$ be increasing in z , or equivalently that $(1-z)c''(z)/c'(z) \geq 2$. Finally, notice that $\tilde{c}'(0) = \frac{1}{2} \tau_v^{-1} c'(0)$, so the critical object—the marginal cost at zero precision—is invariant to this transformation.

¹⁴One could consider an alternative economy in which the noise is given by $u_n \sim \text{Normal}(0, \tau_u^{-1}/n^2)$. In this specification, analogously to our baseline model, write the market clearing condition in levels:

$$\sum_{i=1}^n d_i + n u_n = 0$$

The aggregate noise $n u_n$ has constant size τ_u^{-1} by construction. Dividing the market clearing condition by n and taking the limit, we have

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n d_i + u_n \right) = 0$$

By the law of large numbers, the result is $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n d_i = 0$ almost-surely, i.e., the asymptotic economy is noise-less. Therefore, a way to analyze a competitive model analogous to our baseline model is to study the small-noise limit *after* having derived the equilibrium for each τ_u^{-1} .

Furthermore, the conditional distribution of v given (s_i, p) is

$$(v \mid s_i, p) \sim \text{Normal}(\tilde{\mu}_{vi}, \tilde{\sigma}_{vi}),$$

where $\tilde{\mu}_{vi} := \frac{\tau_v^{-1}}{(\sigma_i - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma_i\beta_u^2\tau_u^{-1}} \left(\beta_u^2\tau_u^{-1}s_i + (\sigma_i - \tau_v^{-1})\beta_v p \right)$

$$\begin{aligned} \tilde{\sigma}_{vi} &:= \tau_v^{-1} - \frac{\tau_v^{-2}}{(\sigma_i - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma_i\beta_u^2\tau_u^{-1}} \begin{pmatrix} 1 \\ \beta_v \end{pmatrix}^\top \begin{pmatrix} \beta_v^2\tau_v^{-1} + \beta_u^2\tau_u^{-1} & -\beta_v\tau_v^{-1} \\ -\beta_v\tau_v^{-1} & \sigma_i \end{pmatrix} \begin{pmatrix} 1 \\ \beta_v \end{pmatrix} \\ &= \tau_v^{-1} \frac{(\sigma_i - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma_i - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma_i\beta_u^2\tau_u^{-1}} \end{aligned}$$

Then, the standard CARA solution for asset demand is

$$d_i = \frac{\tilde{\mu}_{vi} - p}{\gamma \tilde{\sigma}_{vi}}$$

Before continuing, note that if agents collect a symmetric amount of information, $z_i = z^*$ for all i so that $\sigma_i = \sigma^*$, then market clearing implies by the law of large numbers

$$p = \frac{\tau_v^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}} \left(\beta_u^2\tau_u^{-1}v + (\sigma^* - \tau_v^{-1})\beta_v p \right) + \gamma\tau_v^{-1} \frac{(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}} u$$

Rearranging this expression, we have

$$\begin{aligned} p &= \frac{1}{1 - \frac{\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}}} \left[\frac{\tau_v^{-1}\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}} v + \gamma\tau_v^{-1} \frac{(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1}} u \right] \\ &= \frac{1}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v} \left[\tau_v^{-1}\beta_u^2\tau_u^{-1}v + \gamma\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}u \right] \end{aligned}$$

Matching the loadings on v and u with β_v and β_u , respectively, we obtain the system of two equations in the two unknowns (β_v, β_u) :

$$\begin{aligned} \beta_v &= \frac{\tau_v^{-1}\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v} \\ \beta_u &= \frac{\gamma\tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_u^2\tau_u^{-1}}{(\sigma^* - \tau_v^{-1})\beta_v^2\tau_v^{-1} + \sigma^*\beta_u^2\tau_u^{-1} - \tau_v^{-1}(\sigma^* - \tau_v^{-1})\beta_v} \end{aligned}$$

The solution is

$$\begin{aligned}\beta_v &= \frac{\tau_v^{-1}[1 + \gamma^2 \tau_u^{-1}(\sigma^* - \tau_v^{-1})]}{\tau_v^{-1} + \gamma^2 \sigma^* \tau_u^{-1}(\sigma^* - \tau_v^{-1})} \\ \beta_u &= \gamma(\sigma^* - \tau_v^{-1})\beta_v\end{aligned}$$

Thus, we have solved (β_v, β_u) in terms of the symmetric equilibrium information acquisition z^* , or equivalently σ^* . These results so far also match the limiting economy of Hellwig (1980) for the case of identical risk aversions.

D.1 Equilibrium information acquisition

Let us now drop i subscripts everywhere, and use the notation $\sigma(z) := \frac{1}{2}\tau_v^{-1}(z^{-1} + 1)$ for the individual-specific signal variance. To solve the information choice, a la Verrecchia (1982), first compute the ex-ante utility

$$\begin{aligned}V(z) &:= \mathbb{E} \left[\max_d \mathbb{E} [\tilde{V} \mid s, p] \right] \\ &= -\mathbb{E} \left[\exp \left(-\frac{1}{2} \frac{(\tilde{\mu}_v - p)^2}{\tilde{\sigma}_v} + \gamma c(z) \right) \right]\end{aligned}$$

To compute this unconditional expectation, note that $\tilde{\mu}_v$ is linear in (s, p) , which has a normal distribution. The joint density is of (s, p) is

$$\begin{aligned}\varphi(s, p) &= \frac{1}{2\pi} K_0^{1/2} \exp \left[-\frac{1}{2} (k_s s^2 + 2k_{sp} s p + k_p p^2) \right] \\ K_0 &:= \frac{1}{\det(\tilde{\Sigma})} = \frac{1}{(\sigma(z) - \tau_v^{-1})\beta_v^2 \tau_v^{-1} + \sigma(z)\beta_u^2 \tau_u^{-1}} \\ k_s &= \tilde{\Sigma}_{11}^{-1} = K_0(\beta_v^2 \tau_v^{-1} + \beta_u^2 \tau_u^{-1}) \\ k_{sp} &= \tilde{\Sigma}_{12}^{-1} = \tilde{\Sigma}_{21}^{-1} = -K_0 \beta_v \tau_v^{-1} \\ k_p &= \tilde{\Sigma}_{22}^{-1} = K_0 \sigma(z)\end{aligned}$$

Furthermore, write $-\frac{1}{2} \frac{(\tilde{\mu}_v - p)^2}{\tilde{\sigma}_v} = a(b_s s + b_p p)^2$ where we define

$$\begin{aligned}a &:= -\frac{1}{2\tilde{\sigma}_v} = -\frac{1}{2} K_0^{-1} \frac{1}{(\sigma(z) - \tau_v^{-1})\beta_u^2 \tau_v^{-1} \tau_u^{-1}} \\ b_s &:= K_0 \beta_u^2 \tau_u^{-1} \tau_v^{-1} \\ b_p &:= K_0 [(\sigma(z) - \tau_v^{-1})\beta_v \tau_v^{-1} - K_0^{-1}]\end{aligned}$$

Using this expression and the normal density, compute

$$\begin{aligned}\mathbb{E}\left[\exp\left(-\frac{1}{2}\frac{(\tilde{\mu}_v - p)^2}{\tilde{\sigma}_v}\right)\right] &= \mathbb{E}[e^{a(b_s s + b_p p)^2}] \\ &= \frac{1}{2\pi} K_0^{1/2} \iint e^{a(b_s s + b_p p)^2} e^{-\frac{1}{2}(k_s s^2 + 2k_{sp} s p + k_p p^2)} ds dp \\ &= \frac{1}{2\pi} K_0^{1/2} \iint \exp\left[-\frac{1}{2}\left(\begin{smallmatrix} s \\ p \end{smallmatrix}\right)^\top \Omega^{-1} \begin{smallmatrix} s \\ p \end{smallmatrix}\right] ds dp,\end{aligned}$$

where

$$\Omega^{-1} := \begin{pmatrix} k_s - 2ab_s^2 & k_{sp} - 2ab_s b_p \\ k_{sp} - 2ab_s b_p & k_p - 2ab_p^2 \end{pmatrix}$$

Since the integral above involves a normal kernel, we obtain

$$V = -\exp[\gamma c(z)] \frac{K_0^{1/2}}{\det(\Omega^{-1})^{1/2}}$$

After a lengthy amount of algebra, we obtain

$$\Omega^{-1} = \begin{pmatrix} (\sigma(z) - \tau_v^{-1})^{-1} & -(\sigma(z) - \tau_v^{-1})^{-1} \\ -(\sigma(z) - \tau_v^{-1})^{-1} & \tau_v + (\sigma(z) - \tau_v^{-1})^{-1} + \frac{(1-\beta_v)^2}{\beta_u^2 \tau_u^{-1}} \end{pmatrix}$$

Therefore,¹⁵

$$\frac{K_0}{\det(\Omega^{-1})} = \left(\tau_v + (\sigma(z) - \tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2 \tau_u^{-1}}\right)^{-1} \left(\beta_u^2 \tau_u^{-1} + (1 - \beta_v)^2 \tau_v^{-1}\right)^{-1}.$$

It is equivalent to maximize $-\log(-V)$, so we solve

$$\max_z -\gamma c(z) + \frac{1}{2} \log\left(\tau_v + (\sigma(z) - \tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2 \tau_u^{-1}}\right) + \frac{1}{2} \log\left(\beta_u^2 \tau_u^{-1} + (1 - \beta_v)^2 \tau_v^{-1}\right)$$

¹⁵Note that Verrecchia (1982) makes an algebra mistake in deriving his equation (7). Indeed, in deriving his ex-ante value function, he obtains (after translating into our notation)

$$\frac{K_0}{\det(\Omega^{-1})} = \left(\tau_v + (\sigma(z) - \tau_v^{-1})^{-1} + \frac{\beta_v^2}{\beta_u^2 \tau_u^{-1}}\right)^{-1} \frac{1}{\beta_u^2 \tau_u^{-1}}.$$

One can check that the mistake originates in his appendix, where he uses (his notation) $a_3 + a_6 = h_0 + s$ rather than the correct expression $a_3 + a_6 = h_0 + s + \frac{(1-\beta)^2}{\gamma^2 V}$. That said, this mistake is inconsequential, because it is multiplicatively separable from the information choice, as can be seen in the subsequent derivations.

Notice that the final term is irrelevant, given z is absent. Also, at this point let us make the replacement $(\sigma(z) - \tau_v^{-1})^{-1} = 2\tau_v \frac{z}{1-z}$. The FOC for z is

$$\gamma(1-z)^2 c'(z) \left[1 + 2 \frac{z}{1-z} + \frac{\beta_v^2 \tau_v^{-1}}{\beta_u^2 \tau_u^{-1}} \right] \begin{cases} \geq 1, & \text{if } z = 0; \\ = 1, & \text{if } z \in (0, 1); \\ \leq 1, & \text{if } z = 1. \end{cases} \quad (\text{D.1})$$

In a symmetric equilibrium where agents choose z^* , we may use the expressions for $\beta_v/\beta_u = 2\gamma^{-1}\tau_v z^*/(1-z^*)$ to rewrite (D.1) as the equilibrium condition

$$\gamma(1-z^*)^2 c'(z^*) \left[1 + 2 \left(\frac{z^*}{1-z^*} \right) + \frac{4}{\gamma^2 \tau_u^{-1} \tau_v^{-1}} \left(\frac{z^*}{1-z^*} \right)^2 \right] \begin{cases} \geq 1, & \text{if } z^* = 0; \\ = 1, & \text{if } z^* \in (0, 1); \\ \leq 1, & \text{if } z^* = 1. \end{cases} \quad (\text{D.2})$$

again with equality if $z^* > 0$. Under some conditions on the cost function, Verrecchia (1982) proves that a solution exists to this condition, hence an NREE exists with endogenous information acquisition. Notice that $\gamma c'(0) < 1$ suffices to ensure that the solution necessarily satisfies $z^* > 0$.

Let us, for reference, also write the pricing function in terms of z^* rather than the variance σ^* . We have

$$p = \beta_v v + \beta_u u \quad (\text{D.3})$$

where

$$\beta_v = \frac{1 + \frac{\gamma^2}{2} \tau_v^{-1} \tau_u^{-1} \frac{1-z^*}{z^*}}{1 + \frac{\gamma^2}{4} \tau_v^{-1} \tau_u^{-1} \frac{1-z^*}{z^*} \frac{1+z^*}{z^*}}$$

$$\beta_u = \frac{\gamma \tau_v^{-1}}{2} \frac{1-z^*}{z^*} \beta_v$$

In general, this economy cannot feature fully-revealing prices regardless of the marginal cost at zero, $c'(0)$. To see this, imagine holding $\tau_u \in (0, \infty)$, $\tau_v \in (0, \infty)$, and $\gamma > 0$ fixed and vary the information cost function $c(\cdot; \chi)$ with some perturbation parameter χ , which then moves $z^*(\chi)$ in equilibrium. Notice from expression (D.3) that fully-revealing prices obtain if and only if $z^*(\chi) \rightarrow 1$ as we move χ in a manner that reduces the information cost. But from the (D.2), we can easily prove that $z^*(\chi) \rightarrow 1$ requires $c'(1; \chi) \leq \frac{\gamma}{4} \tau_u^{-1} \tau_v^{-1}$ for all χ corresponding to $z^*(\chi)$ close enough to 1. In other words, what matters in an NREE for fully-revealing prices is that each agent obtains a perfectly-informative private signal. This places a strong requirement on the cost function at

$z = 1$, with the shape of the cost function at $z = 0$ being irrelevant. In summary, holding (τ_u, τ_v, γ) fixed, informativeness is bounded above in NREE, even if $c'(0) = 0$.

D.2 Two examples: zero and positive marginal cost at zero

We consider two examples: one with $c'(0) = 0$ and the second with $c'(0) > 0$.

Example 1 (quadratic cost). Consider the cost function

$$c(z) = \frac{\kappa}{2} \left(\frac{z}{1-z} \right)^2. \quad (\text{D.4})$$

This function satisfies $c'(0) = 0$ and therefore is capable of generating infinite price informativeness in our baseline Kyle model in the text. To provide an interpretation, recall that the chosen precision over the signal error ε_i satisfies $s_i = \frac{2}{\tau_v^{-1} 1 - z_i}$. Hence, the function (D.4) represents quadratic costs over this signal error precision.

Evaluating the equilibrium condition (D.2) for this case, we have

$$\gamma \kappa \left[\frac{z^*}{1-z^*} + 2 \left(\frac{z^*}{1-z^*} \right)^2 + \frac{4}{\gamma^2 \tau_u^{-1} \tau_v^{-1}} \left(\frac{z^*}{1-z^*} \right)^3 \right] = 1 \quad (\text{D.5})$$

Thus, $z^*/(1-z^*)$ solves a cubic equation. There is exactly one root satisfying $z^* \in (0, 1)$.

Example 2 (linear cost). Now, consider the cost function¹⁶

$$c(z) = \chi \left(\frac{z}{1-z} \right), \quad \text{where } \gamma \chi < 1. \quad (\text{D.6})$$

This function satisfies $c'(0) = \chi > 0$. (One must take caution interpreting comparative statics with respect to χ , because here χ modulates the *entire cost function* in addition to the marginal cost $c'(0)$.) Also note that information costs are “larger” here than for the quadratic case above. Indeed, since the chosen precision over the signal error ε_i satisfies $s_i = \frac{2}{\tau_v^{-1} 1 - z_i}$, the function (D.6) represents linear costs over signal error precision.

Evaluating the equilibrium condition (D.2) for this case, we have

$$\gamma \chi \left[1 + 2 \left(\frac{z^*}{1-z^*} \right) + \frac{4}{\gamma^2 \tau_u^{-1} \tau_v^{-1}} \left(\frac{z^*}{1-z^*} \right)^2 \right] = 1 \quad (\text{D.7})$$

Thus, $z^*/(1-z^*)$ solves a quadratic equation. Assuming $\gamma \chi < 1$, there exists a unique

¹⁶This cost function is the one which is minimally-convex in signal precisions—see footnote 13.

positive solution z^* . (If $\gamma\chi > 1$, then there is no positive solution, so $z^* = 0$ is the equilibrium.) This unique solution is

$$\frac{z^*}{1 - z^*} = \frac{\gamma\tau_u^{-1}\tau_v^{-1}}{4\chi} \left[-\gamma\chi + \sqrt{(\gamma\chi)^2 + (1 - \gamma\chi)\frac{4\chi}{\gamma\tau_u^{-1}\tau_v^{-1}}} \right] \quad (\text{D.8})$$

D.3 Price informativeness, volatility, and liquidity

Analogous to the baseline model, define the following measures of (relative) price informativeness, volatility, and liquidity:

$$\mathcal{I} := 1 - \frac{\text{Var}[v \mid p]}{\text{Var}[p]}$$

$$\mathcal{V} := \text{Std}[p]$$

$$\mathcal{L} := [\partial p / \partial u]^{-1}$$

The informativeness and volatility measures are exactly analogous to our baseline model. The liquidity measure deserves discussion. Note that $\mathcal{L}^{-1}$ is the price response to an exogenous increase in demand, whereas in our baseline model it was the loading of price on aggregate demand. The definition is written this way for the NREE because “aggregate demand” is not well-defined here—it is zero by market clearing. But a noise shock represents an exogenous increase in asset demand that must be absorbed by the informed traders, and hence generates a price impact. As an alternative way to justify our definition of $\mathcal{L}^{-1}$, recall that price impact in the baseline model comes out exactly equal to the price loading on the noise shock u . That is, the result for price impact in our baseline model coincides with $\partial p / \partial u$. By contrast, volume is not well-defined in the competitive economy, so we do not examine it.

Using expression (D.3) and properties of the joint normal distribution, compute

$$\mathcal{I} = \frac{\beta_v^2 \tau_v^{-1}}{\beta_u^2 \tau_u^{-1} + \beta_v^2 \tau_v^{-1}} = \frac{1}{1 + \frac{\gamma^2}{4} \left(\frac{1 - z^*}{z^*} \right)^2 \tau_u^{-1} \tau_v^{-1}} \quad (\text{D.9})$$

$$\mathcal{V} = \sqrt{\beta_u^2 \tau_u^{-1} + \beta_v^2 \tau_v^{-1}} \quad (\text{D.10})$$

$$\mathcal{L} = \beta_u^{-1} \quad (\text{D.11})$$

D.4 Theoretical results on observables

We have the following proposition characterizing how informativeness, volatility, and liquidity co-move with τ_u^{-1} and τ_v^{-1} .

Proposition D.1. *In the NREE model above, the following hold:*

- (i) *Informativeness $\mathcal{I}$ is always decreasing with fundamental uncertainty τ_v^{-1} and noise τ_u^{-1} .*
- (ii) *Liquidity $\mathcal{L}$ is always decreasing with fundamental uncertainty τ_v^{-1} and increasing with noise τ_u^{-1} .*
- (iii) *Volatility $\mathcal{V}$ is always increasing with fundamental uncertainty τ_v^{-1} . Provided either equilibrium information-collection z^* is not too close to 1 or the curvature of the information cost function is not too high, $\mathcal{V}$ is also increasing with noise τ_u^{-1} .*

Preliminaries. As a preliminary, we use the equilibrium condition (D.2) to examine how information-collection z^* varies with noise and uncertainty. Define the object

$$\nu(z) := (1 - z) \frac{c''(z)}{c'(z)}. \quad (\text{D.12})$$

For the information cost to be convex in precision, we require $\nu(z) \geq 2$ for all z (see footnote 13). Using the implicit function theorem and differentiating (D.2), assuming $z^* \in (0, 1)$, we have

$$\begin{aligned} \frac{d \log z^*}{d \log \tau_u^{-1}} &= \frac{d \log z^*}{d \log \tau_v^{-1}} \\ &= \frac{4z^* \frac{\tau_u \tau_v}{\gamma^2} (1 - z^*)}{4z^* \frac{\tau_u \tau_v}{\gamma^2} (2(1 - z^*) + z^* \nu(z^*)) + (1 + z^*)(1 - z^*) \nu(z^*) - 2z^*(1 - z^*)} \end{aligned} \quad (\text{D.13})$$

Since $\nu(z) \geq 2$, we have that

$$0 < \frac{d \log z^*}{d \log \tau_u^{-1}} = \frac{d \log z^*}{d \log \tau_v^{-1}} \leq \frac{1 - z^*}{2} \frac{4z^* \frac{\tau_u \tau_v}{\gamma^2}}{4z^* \frac{\tau_u \tau_v}{\gamma^2} + 1 - z^*} < \frac{1 - z^*}{2}. \quad (\text{D.14})$$

We make repeated use of (D.14) below.

For the analysis of liquidity and volatility below, let us also note the following useful

relationship. By examining the expressions (D.9)-(D.10)-(D.11), we see that

$$1 - \mathcal{I} = \frac{\tau_u^{-1}}{(\mathcal{L}\mathcal{V})^2}. \quad (\text{D.15})$$

Consequently, it holds that

$$\begin{aligned} \frac{d \log(1 - \mathcal{I})}{d \log \tau_u^{-1}} &= 1 - 2 \left(\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} + \frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} \right) \\ \frac{d \log(1 - \mathcal{I})}{d \log \tau_v^{-1}} &= -2 \left(\frac{d \log \mathcal{L}}{d \log \tau_v^{-1}} + \frac{d \log \mathcal{V}}{d \log \tau_v^{-1}} \right) \end{aligned}$$

Informativeness in NREE. We start by identifying how informativeness varies with τ_u and τ_v in NREE. Using the expression for $\mathcal{I}$ in (D.9), we have that

$$\begin{aligned} \frac{d \log \mathcal{I}}{d \log \tau_u^{-1}} &= \frac{\gamma^2}{2} \frac{1 - z^*}{(z^*)^2} \mathcal{I} \tau_v^{-1} \left[\frac{d \log z^*}{d \log \tau_u^{-1}} - \frac{1 - z^*}{2} \right] \\ \frac{d \log \mathcal{I}}{d \log \tau_v^{-1}} &= \frac{\gamma^2}{2} \frac{1 - z^*}{(z^*)^2} \mathcal{I} \tau_u^{-1} \left[\frac{d \log z^*}{d \log \tau_v^{-1}} - \frac{1 - z^*}{2} \right] \end{aligned}$$

Using the upper bound in (D.14), we have that whenever $z^* \in (0, 1)$ it holds that

$$\frac{d \log \mathcal{I}}{d \log \tau_u^{-1}} < 0 \quad \text{and} \quad \frac{d \log \mathcal{I}}{d \log \tau_v^{-1}} < 0$$

as desired. ■

Liquidity in NREE. We now analyze how liquidity $\mathcal{L}$ varies with noise τ_u^{-1} and fundamental uncertainty τ_v^{-1} in NREE.

First, substitute the expression for β_u from (D.3) into the expression for liquidity in (D.11) and simplify terms to write

$$\mathcal{L} = \underbrace{\left(\frac{D_2}{D_1} \right)}_{\beta_v^{-1}} \left(\frac{\gamma}{2} \frac{1 - z^*}{z^*} \tau_v^{-1} \right)^{-1}$$

where $D_1 := 1 + \frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1 - z^*}{z^*}$, $D_2 := 1 + \frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1 - z^*}{z^*} \frac{1 + z^*}{2z^*}$.

We first differentiate the β_v term. We have

$$\frac{d \log \beta_v}{d \log \tau_u^{-1}} = \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \left[1 + \frac{d \log(\frac{1-z^*}{z^*})}{d \log \tau_u^{-1}} \right]}{D_1} - \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \frac{1+z^*}{2z^*} \left[1 + \frac{d \log(\frac{1-z^*}{z^*} \frac{1+z^*}{2z^*})}{d \log \tau_u^{-1}} \right]}{D_2}$$

We compute

$$\begin{aligned} \frac{d \log(\frac{1-z^*}{z^*})}{d \log \tau_u^{-1}} &= -\frac{1}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \\ \frac{d \log(\frac{1+z^*}{2z^*})}{d \log \tau_u^{-1}} &= -\frac{1}{1+z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \end{aligned}$$

Substituting and doing some algebra, we have

$$\frac{d \log \beta_v}{d \log \tau_u^{-1}} = \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*}}{D_1 D_2} \left[-\frac{1-z^*}{2z^*} + \frac{1}{z^*} \left(1 + \frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \right) \frac{d \log z^*}{d \log \tau_u^{-1}} \right] \quad (\text{D.16})$$

Therefore,

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} = \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*}}{D_1 D_2} \left[\frac{1-z^*}{2z^*} - \frac{1}{z^*} \left(1 + \frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \right) \frac{d \log z^*}{d \log \tau_u^{-1}} \right] + \frac{1}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}}$$

Combining terms, and simplifying the expression, we have

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} = \frac{1}{D_1 D_2 (z^*)^2} \left[\frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} (1-z^*)^2 + \frac{F(z^*)}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \right],$$

where

$$F(z) := z^2 - \frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} (1-z)(1-5z) + \left(\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \right)^2 (1-z)^2$$

Using the fact that $\frac{d \log z^*}{d \log \tau_u^{-1}} > 0$, we have

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > \frac{\frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1}}{D_1 D_2 (z^*)^2} \left[(1-z^*)^2 + G(z^*) \frac{d \log z^*}{d \log \tau_u^{-1}} \right],$$

where

$$G(z) := \gamma^2 \tau_u^{-1} \tau_v^{-1} (1 - z) - (1 - 5z)$$

There are two cases to consider. If $\gamma^2 \tau_u^{-1} \tau_v^{-1} \geq 1$, then $G(z) > 0$ for all $z \in (0, 1)$, and so the fact that $\frac{d \log z^*}{d \log \tau_u^{-1}} > 0$ implies $\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > 0$.

If instead $\gamma^2 \tau_u^{-1} \tau_v^{-1} < 1$, then $G(0) < 0$, and so we proceed as follows. We substitute $\frac{d \log z^*}{d \log \tau_u^{-1}}$ from (D.13), and use the fact that $\nu(z) \geq 2$ to get the following bound that applies when $G(z^*) < 0$:

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > \frac{\frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1}}{D_1 D_2 (z^*)^2} \left[(1 - z^*)^2 + \frac{G(z^*)}{2} \frac{4z^*(1 - z^*) \frac{\tau_u \tau_v}{\gamma^2}}{4z^* \frac{\tau_u \tau_v}{\gamma^2} + (1 - z^*)} \right], \quad \text{if } G(z^*) < 0.$$

After some algebra, we can rewrite this bound as

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > \frac{\frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} (1 - z^*)}{D_1 D_2 (z^*)^2} \left[1 + z^* - \frac{1}{2} \frac{4z^*(1 - z^*) \frac{\tau_u \tau_v}{\gamma^2}}{4z^* \frac{\tau_u \tau_v}{\gamma^2} + (1 - z^*)} \right], \quad \text{if } G(z^*) < 0.$$

It is easy to see that the term in square brackets is positive, proving that $\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > 0$ in this second case as well. Thus, we have in general

$$\frac{d \log \mathcal{L}}{d \log \tau_u^{-1}} > 0.$$

For the effect of fundamental uncertainty, the proof is simpler. In the next section, we prove that $\frac{d \log \mathcal{V}}{d \log \tau_v^{-1}} > 0$. Given that we also previously showed $\frac{d \log \mathcal{I}}{d \log \tau_v^{-1}} < 0$, equation (D.15) implies that

$$\frac{d \log \mathcal{L}}{d \log \tau_v^{-1}} < 0$$

as desired. ■

Volatility in NREE. Finally, we analyze how volatility $\mathcal{V}$ varies with noise τ_u^{-1} and fundamental uncertainty τ_v^{-1} .

Substitute the expressions for β_u, β_v from (D.3) into the expression for volatility in

(D.10) in order to rewrite volatility as

$$\mathcal{V} = \underbrace{\left(\frac{D_1}{D_2}\right)}_{=\beta_v} \tau_v^{-1/2} \left[1 + \tau_u^{-1} \tau_v^{-1} \left(\frac{\gamma}{2} \frac{1-z^*}{z^*}\right)^2\right]^{1/2}$$

$$\text{where } D_1 := 1 + \frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*}, \quad D_2 := 1 + \frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \frac{1+z^*}{2z^*}.$$

For reference, we also define the additional variable

$$D_3 := 1 + \tau_u^{-1} \tau_v^{-1} \left(\frac{\gamma}{2} \frac{1-z^*}{z^*}\right)^2$$

We start with noise τ_u^{-1} . Differentiating, using (D.16), and simplifying the result, we have

$$\begin{aligned} \frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} &= \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \left(\frac{1-z^*}{z^*}\right)^2}{D_1 D_2 D_3} \times \\ &\quad \left[\frac{D_1 D_2 - 2D_3}{4} \left(1 - \frac{2}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}}\right) + \frac{D_3}{z^*} \frac{\gamma^2}{4} \tau_u^{-1} \tau_v^{-1} \frac{d \log z^*}{d \log \tau_u^{-1}} \right] \end{aligned}$$

There are two cases to consider. In the first case, if z^* is not too close to 1, then¹⁷

$$D_1 D_2 - 2D_3 = -1 + \gamma^2 \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} + \left(\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*}\right)^2 \frac{1+z^*}{2z^*}$$

is positive, which implies by the bounds (D.14) that $\frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} > 0$. In the second case, suppose z^* is close to 1. We rewrite the derivative as

$$\begin{aligned} \frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} &= \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \left(\frac{1-z^*}{z^*}\right)^2}{D_1 D_2 D_3} \times \left[\frac{D_1 D_2 - D_3}{4} \left(1 - \frac{2}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}}\right) \right. \\ &\quad \left. - \frac{D_3}{4} \left(1 - \frac{2}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}}\right) + \frac{D_3}{4} \frac{\gamma^2 \tau_u^{-1} \tau_v^{-1}}{z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \right] \end{aligned}$$

¹⁷A sufficient condition for $D_1 D_2 - 2D_3 > 0$ is that

$$z^* < \frac{\gamma^2 \tau_u^{-1} \tau_v^{-1}}{2(\sqrt{2}-1) + \gamma^2 \tau_u^{-1} \tau_v^{-1}}.$$

Note that

$$D_1 D_2 - D_3 = \frac{1+z^*}{2z^*} \left(\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \right)^2 + \gamma^2 \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*}$$

is strictly positive, and so

$$\frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} > \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \left(\frac{1-z^*}{z^*} \right)^2}{4D_1 D_2} \left[\frac{\gamma^2 \tau_u^{-1} \tau_v^{-1}}{z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} - \left(1 - \frac{2}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \right) \right] \quad (\text{D.17})$$

Now, plugging in $\frac{d \log z^*}{d \log \tau_u^{-1}}$ from (D.13), the term in square brackets of the above expression can be written:

$$\begin{aligned} & \frac{\gamma^2 \tau_u^{-1} \tau_v^{-1}}{z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} - \left(1 - \frac{2}{1-z^*} \frac{d \log z^*}{d \log \tau_u^{-1}} \right) \\ &= \frac{(1-z^*) \left[2 - \frac{\nu(z^*)}{2} \right] - [4z^* \frac{\tau_u \tau_v}{\gamma^2} + 1 - z^*] z^* \left(\frac{\nu(z^*)}{2} - 1 \right)}{4z^* \frac{\tau_u \tau_v}{\gamma^2} \left((1-z^*) + z^* \frac{\nu(z^*)}{2} \right) + (1-z^*) \frac{\nu(z^*)}{2} + z^* (1-z^*) \left(\frac{\nu(z^*)}{2} - 1 \right)} \end{aligned}$$

For $\nu(z^*)$ close enough to its lower-bound of 2, this expression is also strictly positive. Thus, we have proved that

$$\frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} > 0$$

for any parameters such that either (i) z^* is far enough from its upper-bound of 1; or (ii) $\nu(z^*)$ is close enough to its lower-bound of 2.

Regarding fundamental uncertainty, since $\frac{d \log z^*}{d \log \tau_u^{-1}} = \frac{d \log z^*}{d \log \tau_v^{-1}}$, we have

$$\frac{d \log \mathcal{V}}{d \log \tau_v^{-1}} = \frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} + \frac{1}{2} \quad (\text{D.18})$$

Using (D.17) and using $\frac{d \log z^*}{d \log \tau_u^{-1}} > 0$, we have that

$$\frac{d \log \mathcal{V}}{d \log \tau_u^{-1}} > - \frac{\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \left(\frac{1-z^*}{z^*} \right)^2}{4 + 4 \frac{1+z^*}{2z^*} \left(\frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \right)^2 + 2 \frac{\gamma^2}{2} \tau_u^{-1} \tau_v^{-1} \frac{1-z^*}{z^*} \frac{1-z^*+4z^*}{z^*}} > -\frac{1}{2}$$

Combining this result with (D.18), we have that

$$\frac{d \log \mathcal{V}}{d \log \tau_v^{-1}} > 0$$

as desired. ■

D.5 Fully-revealing prices in two limiting cases

Under what conditions can prices be fully-revealing in an NREE? We show below that fully-revealing prices can occur in two limits, $\tau_u^{-1} \rightarrow 0$ and $\gamma \rightarrow 0$. The small-noise limit bears a closer analogy to our main results, in the sense that a fully-revealing equilibrium can emerge if $c'(0) = 0$ but not if $c'(0) > 0$. By contrast, the risk-neutral limit always features a fully-revealing equilibrium.

Small-noise limit. As a first result, notice that as noise vanishes ($\tau_u^{-1} \rightarrow 0$), the left-hand-side of (D.2) blows up for any $z^* > 0$. Hence, a requirement is $z^* \rightarrow 0$ as $\tau_u^{-1} \rightarrow 0$, assuming an equilibrium exists for each τ_u^{-1} along this sequence.¹⁸ This proves right away that, in a competitive model, individual information collection requires the noise to be *growing* with the number of traders, such that per-capita noise remains non-trivial. This is why the literature universally adopts this setup. However, it is still enlightening to examine the small-noise limit.

Although information-collection vanishes individually in this small-noise limit, so does per-capita noise, and so it is theoretically possible that the price becomes informative. In other words, it becomes a delicate balance of limits. Let us consider the case $0 < \gamma c'(0) < 1$, which ensures that $z^* > 0$ along the sequence of equilibria (if $\gamma c'(0) \geq 1$, then z^* would just collapse to zero for any small enough noise). We will consider the case $c'(0) = 0$ afterward.

If $0 < \gamma c'(0) < 1$, inspection of condition (D.2) shows that $\tau_u^{-1}/(z^*)^2$ must not explode as $\tau_u^{-1} \rightarrow 0$. We can also show that $\tau_u^{-1}/(z^*)^2$ cannot be vanishing, because if it did, the left-hand-side of (D.2) would necessarily exceed 1 for all small enough τ_u^{-1} (given we know $z^* > 0$ along the sequence). Hence, $\tau_u^{-1}/(z^*)^2$ has a finite, non-zero limit. Now, returning to the equilibrium price in (D.3), and using this fact, we find that β_v and $\tau_u^{-1/2}\beta_u$ both converge to finite constants. The limiting value for β_v is necessarily positive but less than 1. The limiting value for $\tau_u^{-1/2}\beta_u$ is also necessarily positive and

¹⁸We conjecture this existence is in fact true. Studying the equilibrium condition (D.2), all that is required is for z^* to vanish at the order of $\tau_u^{-1/2}$ if $c'(0) > 0$, or potentially at any slower rate if $c'(0) = 0$.

finite. Therefore, we have proven that the small-noise limit, if such equilibrium exists, must still have a noisy price. In particular, prices can never be fully revealing.

If $c'(0) = 0$, the analysis is more delicate. Suppose $z^* \sim A\tau_u^{-\zeta/2}$ for some $A > 0$ and $\zeta > 0$. The left-hand-side of (D.2) behaves asymptotically like

$$\frac{4A^2}{\gamma\tau_v^{-1}}c'(A\tau_u^{-\zeta/2})\tau_u^{1-\zeta} + o(\tau_u^{-1})$$

For this expression to remain non-trivial, hence remain consistent with (D.2), we see that $\zeta < 1$ is required. In that case, $\beta_v \rightarrow 1$ and $\tau_u^{-1/2}\beta_u \rightarrow 0$, so that $p \rightarrow v$. Thus, assuming an equilibrium exists along this sequence, the economy approaches one with a fully-revealing price and nevertheless traders acquire information all along the sequence.

To develop an understanding, let's assume that $c(z) = \frac{\kappa}{1+\rho}z^{1+\rho}$ for some $\rho > 0$. Then, $c'(z) = \kappa z^\rho$, so

$$\frac{4A^2}{\gamma\tau_v^{-1}}c'(A\tau_u^{-\zeta/2})\tau_u^{1-\zeta} = \frac{4A^{2+\rho}\kappa}{\gamma\tau_v^{-1}}\tau_u^{-(\zeta-1+\zeta\rho/2)}$$

This expression needs to remain non-zero and finite, so we need $\zeta = (1 + \rho/2)^{-1}$, which is appropriately below 1. This shows that there exists a sequence $z^*(\tau_u^{-1})$ vanishing at rate $\zeta = (1 + \rho/2)^{-1}$ such that the equilibrium condition (D.2) holds with equality for every τ_u^{-1} small enough. Consequently, equilibria with information-acquisition exist for all τ_u^{-1} small enough, and these converge to a fully-revealing equilibrium.

Risk-neutral limit. Second, consider what happens if agents become asymptotically risk-neutral, $\gamma \rightarrow 0$. Going back to the equilibrium price, and assuming any information is gathered at all (i.e., z^* converges to a non-zero limit), this makes the price become fully-revealing, i.e., noise-free with $\beta_u \rightarrow 0$ and $\beta_v \rightarrow 1$. However, the left-hand-side of equilibrium condition (D.2) explodes unless $z^* \rightarrow 0$. Thus, a requirement is $z^* \rightarrow 0$ as $\gamma \rightarrow 0$.

How fast does $z^* \rightarrow 0$? As a first possibility, assume $z^* = 0$ for γ sufficiently close to zero, i.e., information vanishes before risk aversion does. This cannot be an equilibrium, because for γ sufficiently small, equilibrium condition (D.2) will also be smaller than 1, a contradiction.

As a second possibility, suppose $z^* \sim A\gamma^\zeta$ as $\gamma \rightarrow 0$, with some $A > 0$ and some $\zeta > 0$. Note that the left-hand-side of equilibrium condition (D.2) asymptotically looks

like

$$\frac{4A^2}{\tau_u^{-1}\tau_v^{-1}}c'(0)\gamma^{2\zeta-1} + o(\gamma)$$

If $\zeta > 1/2$, then this expression vanishes in the $\gamma \rightarrow 0$ limit, which from (D.2) implies $z^* = 1$ eventually, contradicting the fact that $z^* \rightarrow 0$. Therefore, the appropriate rate of convergence for z^* is $\zeta \leq 1/2$. There are two cases. If $c'(0) > 0$, then clearly $\zeta = 1/2$ is required. If $c'(0) = 0$, then $\zeta < 1/2$ is required, with the exact rate of convergence determined by c'' near zero. In either case, the fact that $\zeta \leq 1/2$ means that $\beta_v \rightarrow 1$ and $\beta_u \rightarrow 0$ as $\gamma \rightarrow 0$. Therefore, we have proven that as γ vanishes, equilibrium becomes fully-revealing, if it exists.